bb-buck — Design Document

ai-ee v1 pipeline

generated 2026-08-16 04:40:44

Contents

1 Board and Run Metadata

Field	Value
board	bb-buck
workspace	`boards/bb-buck`
phase	P8
gate status	no gates recorded yet
state created	2026-08-15T15:58:13
state updated	2026-08-16T04:40:43

2 Overview

Brief - bb-buck (as given by the owner, verbatim)

ultra bare bones design: a switch-mode step-down converter. Input 24 V DC nominal (18-30 V) on a screw terminal; output 5 V at up to 2 A on a screw terminal. Bench supply in, resistive load out. Workspace boards/bb-buck.

Orchestrator notes (not owner words)

- Mode token present: `ultra bare bones design:` -> the ultra-bare-bones scope contract in `reference/build-modes.md` applies.
- Block under study: the synchronous/asynchronous step-down (buck) converter.
- Bench-only article: bench DC supply on the input, resistive load on the output.

Stackup

Chosen stackup: **‘JLC2313_1.6‘** - JLCPCB standard 2-layer, 1.6 mm, **1 oz outer copper**,

3 Requirements

Requirements: bb-buck

1. Function

A bare step-down (buck) DC-DC converter, built as a study article. It takes 18-30 V DC (24 V nominal) from a bench power supply on a 2-pin screw terminal and produces 5 V at up to 2 A (10 W) on a second 2-pin screw terminal feeding a resistive load. Non-isolated, common ground in to out. Nothing else is on the board: no MCU, no second rail, no digital interface, no indicators.

BUILD MODE: **ultra-bare-bones** (declared by the brief's opening token, recorded as a P0 decision; contract in `reference/build-modes.md`). The block under study is the buck converter itself. Scope is that block plus exactly the support parts its datasheet requires at the stated operating point, one input and one output interface, and what the fab needs to build the board and the bench needs to hold it. Protection, filtering, indicators, spare rails and enclosure features are OUT by mode, not by engineering judgment. The mode's defaults are applied below (quantity 5, JLC PCBA single-sided, bench 0-50 C indoor, no enclosure, fewest honest layers, smallest honest outline, stop at P9) instead of being asked.

Synchronous vs asynchronous rectification is deliberately NOT fixed here. The brief leaves it open and it is a P1 architecture call, not a requirement. Both are in scope as long as the block meets the stated operating point.

2. Interfaces

Exactly two, both stated by the brief.

- Power input (J1): 2-pin screw terminal (stated). DC only, 18-30 V. ASSUMED: through-hole 2-pole terminal block, 5.08 mm pitch, rated >= 300 V and >= 10 A so it is never the limiting element against the ~0.7 A worst-case input current. Pitch was not stated; 5.08 mm is the common, well-stocked JLC/LCSC choice and accepts bench-supply lead wire comfortably. Polarity silkscreen-marked (+ / -). NOTE: the board has no reverse-polarity protection by mode, so the silk marking is the only defense against a swapped supply - see section 8.
- Power output (J2): 2-pin screw terminal (stated). 5 V, up to 2 A. ASSUMED: same terminal family and part as J1 for BOM consolidation, placed on a different board edge and silk-labelled distinctly (VIN vs 5V OUT) so the two cannot be confused at the bench.
- No other external interface exists or is wanted: no USB, no header, no comms bus, no external enable, no power-good, no fault output, no LED. All excluded by mode.
- Measurement access ANSWERED (A4): exactly one switch-node probe pad with an adjacent ground pad. No other test points - VIN, VOUT and GND are reachable at the terminals.

3. Power

- Source: bench DC power supply, 18-30 V, 24 V nominal (STATED explicitly by the brief - not a guess, not a battery, not mains-derived on this board, not automotive). Section 8 depends on this being stated.
- Load: resistive, up to 2 A at 5 V (STATED). ASSUMED: the load range is 0 to 2 A, i.e. the board must also be stable and safe with NO load connected (the realistic bench case of powering up before the resistor is attached). This is the conservative reading of "up to 2 A" and it constrains the converter's light-load behavior, so it is recorded rather than left silent.

- ASSUMED: 2 A is the absolute maximum output, continuous. A resistive load draws no surge beyond its steady value, so no peak allowance above 2 A is carried. Inrush into the board's own output capacitance at startup is the converter's soft-start problem, not a load requirement.
- Output rail: one rail, fixed 5 V nominal. ASSUMED: fixed output, not adjustable, no trim, no sequencing. Accuracy and ripple ANSWERED (A3): +/-3 % DC (4.85-5.15 V) across the full 18-30 V input and 0-2 A load range, ripple <= 50 mV peak-to-peak.
- Input current (GUESS, for sizing only): 10 W out at an assumed 88-93 % efficiency -> 10.8-11.4 W in. At 18 V low line ~0.60-0.63 A; at 24 V ~0.45-0.47 A; at 30 V high line ~0.36-0.38 A. Size the input terminal, input copper and input capacitor RMS rating for >= 1 A continuous with a switched peak of ~2.5 A. Output terminal and copper for >= 2.5 A continuous.
- Board dissipation (GUESS): ~0.8-1.4 W typical, ~1.8 W at a pessimistic 85 % efficiency. Worst case sits at the 30 V high-line corner where switching loss dominates. At the mode's 50 C maximum ambient with natural convection this is a real thermal driver on a small outline - copper area under the converter is a design input, not an afterthought.
- Duty cycle (derived, for P1): D ~ Vout/Vin = 0.28 at 18 V, 0.21 at 24 V, 0.17 at 30 V. The 30 V corner sets the minimum on-time constraint, which bounds how high the switching frequency can go for a given part - a real P1/P2 selection constraint, flagged here, not decided here.
- No battery, no charging, no energy storage function, no backup rail. None stated and none implied by a bench supply source.
- Excluded by mode (SCOPE decisions, not engineering conclusions - do not record any of these as generally unnecessary): input TVS/clamp, reverse polarity protection, input fuse, pi/EMI input filter, OVP/OCP/UVLO beyond what lives inside the converter IC, output LC filter beyond what the datasheet requires.

4. Environment

Mode defaults - applied, not asked.

- Ambient: 0 to 50 C, indoor bench.
- Cooling: natural convection only. No forced airflow, no heatsink, no chassis thermal path.
- No enclosure. No ingress (IP) rating. No vibration or shock requirement. No humidity requirement. No conformal coating.
- No formal EMC/emissions campaign - this is a bench study article, not a product. Tight power-loop layout is still required because it is what makes the block correct, not because a test demands it.

5. Size & mounting

- Outline: **no HARD cap.** Nothing here binds permanently at P5 board_init. The mode governs instead: the smallest outline that keeps the layout HONEST - never so tight that the power loop, the copper thermal area or the terminal spacing stop being representative of a real 24 V -> 5 V / 2 A buck, and never padded for features this board does not have.
- Non-binding expectation for planning only (NOT a cap): roughly 30-40 mm x 20-28 mm, dominated by the two screw terminals plus the converter, inductor and capacitors. If the honest layout needs more, it takes more.

- Layers: ASSUMED 2, per the mode's "fewest layers the block honestly needs". If the P2/P4 thermal work shows 2 layers cannot carry ~1.2-1.8 W at 50 C ambient inside the honest outline, 4 layers is allowed under the same rule. Not fixed here either way.
- Mounting: ASSUMED 4 x M3 clearance holes (3.2 mm) inset from the corners, so the board can sit on standoffs at the bench. Mounting the bare board is explicitly in mode scope. If the honest outline turns out too small for four, two on opposite corners is acceptable.
- Height: ASSUMED no maximum. The board is open on a bench with no enclosure, and the screw terminals themselves stand ~10-12 mm tall with wire fitted, so a height cap would be arbitrary.
- Connector placement: ASSUMED both terminals on board edges with their wire openings facing outward, on different edges, so bench wiring does not cross the board. Which edges is the layout engineer's call.

6. Quantity & budget

Mode defaults - applied, not asked.

- Build quantity: 5.
- Cost: minimal, with no hard per-unit cap. Prefer JLC Basic/Preferred parts wherever a Basic part actually meets the electrical requirement; Extended is allowed for the converter IC and the power inductor, where Basic stock is thin in this class.

7. Assembly

Mode defaults - applied, not asked.

- JLCPCB PCBA, single-sided: all SMT parts on the top side only. The bottom side stays clear of SMT so it can serve as unbroken thermal/return copper.
- The two screw terminals are through-hole. ASSUMED: fitted by JLCPCB through-hole assembly if offered at acceptable cost for the chosen part, hand-soldered after SMT otherwise. Either path leaves the schematic and the footprint unchanged.
- Pipeline stops at P9 (fab package + DFM). Ordering is a separate owner decision and is not requested here.

8. Compliance/safety flags

No safety question is open on this board, and that conclusion is derived, not assumed. The mode grants no silence here - it simply happens that the brief states the two facts that would otherwise have to be asked.

- **Mains voltage: DOES NOT APPLY.** The board sees DC only. Generating that DC is out of scope; the brief names a bench supply as the source.
- **Batteries: DOES NOT APPLY.** The source is STATED as a bench power supply. This is the one question that has no default and must never be guessed, and the brief answers it explicitly, so it is not asked. No battery on the board, no charging function, no pack chemistry, no deep-discharge or UVLO behavior required. If the owner ever runs this board from a battery instead, that is a new requirement and this section must be re-opened.
- **Motors: DOES NOT APPLY.** The load is STATED as resistive. No inductive, stalling or regenerative load, therefore no reverse energy into the output and no startup surge beyond the resistor's own draw.

- **RF transmit: DOES NOT APPLY.** No radio function.
- **High current (>3 A): DOES NOT APPLY.** Maximum continuous current anywhere on the board is the 2 A output. Input current peaks at ~0.63 A continuous at the 18 V low-line corner. The largest instantaneous current is the inductor peak, GUESSED at ~2.3-2.6 A depending on the ripple ratio P1 chooses - still under 3 A. The threshold is not approached closely enough to be a judgment call.
- **>30 V: DOES NOT APPLY at the stated number, but the range's top edge is genuinely under-defined.** Reasoning, explicitly:
  - The stated maximum is 30 V, and the threshold is *greater than* 30 V. 30 is not >30, so the flag does not trip on the stated figure. This is a boundary case, so it is reasoned about rather than waved past.
  - No real hazard exists at 30 V DC. It is well below the 60 V DC limit for a non-hazardous DC source under IEC 62368-1 (ES1), so there is no shock hazard to a person handling the board. Fault energy is bounded by the bench supply's own current limit, and at <= 1 A input there is no arc or energy hazard at the terminals. No safety question follows from the voltage itself.
  - Downstream consequence worth noting: at exactly 30 V, the VIN-to-GND net pair is 30 V apart, which the pipeline's `check_creepage` treats as a clean no-op (its derived check engages only above 30 V). If the answer to Open question 1 lifts the ceiling above 30 V, that check engages for real and IPC-2221 spacing starts binding the layout. The boundary is therefore not cosmetic.
  - What WAS genuinely ambiguous - whether 30 V is the *operating* or the *survival* ceiling, and what the input does on a live connection - is now ANSWERED (A1, A2): 30 V is a hard maximum OPERATING input, and there is no live hot-plug. The converter carries >= 36 V absolute-max as its own headroom; no clamp is added. The flag therefore stays DOES-NOT-APPLY on settled requirements, not merely on the boundary reading.
- **Reverse polarity: noted, not flagged.** The mode excludes reverse polarity protection, so connecting the bench supply backwards will destroy the converter. This is an accepted consequence of the declared scope on a bench article with silk-marked polarity, not a safety hazard to a person, and not a finding for a reviewer to raise. It is recorded so the consequence is visible rather than implicit.

9. Open questions - ALL FOUR ANSWERED 2026-08-15 (owner)

STATUS: **closed.** The owner took the stated DEFAULT on all four. Each answer is recorded as a P0 `state.py decision` and is now a REQUIREMENT, not a question. P1/P2 read the ANSWERED block below; the original question text is kept underneath it for provenance only.

Answers (binding)

- **A1 - Input ceiling.** 30 V is a HARD maximum operating input. Choose a converter whose absolute-maximum input rating is >= 36 V class (~20 % headroom above 30 V). The headroom lives in the PART, not in added components - no clamp, no TVS. Consequence retained: VIN-to-GND stays at or below 30 V, so `check_creepage` remains a no-op and IPC-2221 spacing does not bind the layout.
- **A2 - No live hot-plug.** The input is never connected while the supply is live: wires land first, then the supply ramps from zero. The ~2x lead-inductance ringing case (~60 V from a 30 V setting) is therefore OUT of the requirement set. The datasheet-required bulk input capacitance stays fitted and damps the incidental case; no damping component is added. A 36 V-class part is sufficient under this answer.

- **A3 - Output spec. AMENDED TWICE AT P3/P4 - read the amendment below, not just this paragraph.** As originally answered: +/-3 % DC, i.e. 4.85-5.15 V, held across the FULL 18-30 V input range and the FULL 0-2 A load range, with <= 50 mV peak-to-peak output ripple. **A3 AS AMENDED (this is the binding version).** A3 now has a TWO-REGION shape, and both amendments have the same root cause: the chosen part runs PFM / diode-emulation at light load and has no MODE pin to defeat it.

Load	DC accuracy	Ripple
~200 mA to 2 A	+/-3 % (4.85-5.15 V) BINDING	<= 50 mV pk-pk BINDING
0 to ~200 mA	PFM light-load rise ACCEPTED	PFM burst ripple ACCEPTED

- *Amendment 1 (P3, owner-ruled).* The DDA package has no MODE/FCCM pin and is auto-mode only (datasheet 8.4.1, which states outright that PFM "yields larger output voltage ripple"). None of the three accept criteria
  - forced-CCM, a strappable MODE pin, or a datasheet figure at this operating point - was satisfiable. The mode excludes the usual fix (a preload resistor is excluded conditioning), so the owner relaxed ripple below ~200 mA rather than override the mode or re-source the part.
  - *Amendment 2 (P4, owner-ruled).* The SAME PFM behaviour also raises the output VOLTAGE at light load: datasheet Sec 7.7 specs regulation as -1.5/+1.5 % for IOUT >= 1 A but **-1.5/+2.5 % for IOUT 0 A to max**, a +1.0 % adder confirmed by Fig 9-19. Stacked on the worst 0-50 C divider corner that gave 5.161 V against the 5.15 V ceiling - outside. So the carve-out extends to DC accuracy below ~200 mA. An earlier statement that DC accuracy was unaffected at all loads was WRONG and is retracted here.
  - *Paid for, not merely relaxed.* The owner took the recentring remedy as well: the feedback divider moved from 100k/24.9k (5.016 V nominal, TI's own Table 9-2 pair) to **102k/25.5k**, an exact 4.000 ratio giving **5.000 V nominal**. That moved the stacked 0 A worst case from 5.161 V to 5.144 V - inside the original window even though it no longer has to be. Both parts are E96, 0.1 %/25 ppm, same Yageo RT family (TCR tracking preserved). The P8 simulation bounds encode this: error bounds stay on the full 4.85-5.15 V window (the divider ratio is load-independent, so the carve-out relaxes the CONVERTER, not the divider), plus a warning-severity bound carrying the +1.0 % light-load stack so it stays visible in the gate. That warning bound is the ONLY check that catches a revert to 100k/24.9k.
- **A4 - Probe access.** Exactly ONE switch-node probe pad with an adjacent ground pad. Nothing else: VIN, VOUT and GND are already reachable at the two screw terminals. Mode-boundary ruling by the owner - the switch node counts as the block's own measurement need on a study article. Keep the pad minimal and inside the switch-node copper that already exists, so it adds as little area as possible to the noisiest node.

Original question text (provenance) 1. **Is 30 V truly the highest the supply will ever be set to, or should the board survive a bit more?** A bench supply set to "30 V" may actually sit at 30.5 V, and a knob can be nudged. This board has no input clamp by design, so the converter chip's own maximum rating is the only thing standing between an overshoot and a dead board. DEFAULT: 30 V is a hard maximum operating input, and the converter is chosen with an absolute-maximum input rating at least ~20 % above it (36 V class or better) purely as selection headroom. No extra parts are added. 2. **Will you ever connect the input wires while the bench supply is already switched on and at voltage?** Connecting live through a metre of lead wire makes the input voltage ring up to roughly twice the supply setting for a few microseconds - about 60 V from a 30 V supply - because the lead inductance rings against the board's ceramic input capacitors. That transient is well beyond what a 36-40 V part tolerates, and the mode excludes the clamp that would normally absorb it, so the answer changes which parts are even eligible. DEFAULT: no live hot-plug. You connect the wires first, then bring the supply up from zero. The bulk input capacitance the datasheet already requires stays fitted and damps the incidental case; nothing extra is added. If you answer "yes, I will hot-plug it", say so - it will force either a much higher-voltage part or a damping component the mode would otherwise exclude, and that is an owner call. 3. **How tight does the 5 V output need to be?** Two numbers: how far from exactly 5.00 V is acceptable, and how much high-frequency ripple can ride on top of it. DEFAULT: +/-3 % DC (4.85-5.15 V) across the full 18-30 V input range and the full 0-2 A load range, and <= 50 mV peak-to-peak output ripple. Ordinary, honest numbers for a study article - not a precision rail, not a loose one. 4. **Do you want one probe point on the switching node?** That means a small pad on the switch node with a ground pad right beside it, so a scope probe with a short ground can measure switch-node ringing and efficiency properly at bring-up. This is the one place the mode's boundary needs an owner ruling: the mode excludes test points "beyond the block's own measurement need", and for a board whose purpose is to be studied - and whose bring-up evidence is what proves out the knowledge records it exists to seed - that need is arguable either way. The tradeoff is real: a pad hangs extra copper on the noisiest node on the board. DEFAULT: yes - exactly one switch-node probe pad with an adjacent ground pad, and nothing else. VIN, VOUT and GND are already reachable at the two screw terminals, so no other test points are added.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Build mode: ultra-bare-bones (token 'ultra bare bones design:' opens the brief)	reference/build-modes.md scope contract: ONE functional block (24V->5V/2A buck) plus only datasheet-required support; one input + one output interface (screw terminals); no protection/filtering/indicators/test points beyond the block's own need; defaults = fewest honest layers, JLC PCBA single-sided, qty 5, bench 0-50C, stop at P9. Research mandatory: first coverage call of each phase runs –maturity-floor proven –research-provisional.
P0	Input ceiling: 30 V is a HARD maximum operating input; converter chosen with Vin absolute-max >= 36 V class (~20% headroom) as selection headroom only - no added clamp parts	Owner answered P0 Q1 with the default. Keeps VIN-GND at/below 30 V so check_creepage stays a no-op and IPC-2221 spacing does not bind the layout; the mode excludes an input clamp so the IC rating is the only defense.
P0	No live hot-plug of the input: wires land first, supply ramps from zero	Owner answered P0 Q2 with the default. Removes the ~2x lead-inductance ringing case (~60 V from 30 V) from the requirement set; datasheet-required bulk input capacitance damps the incidental case, no damping component added.
P0	Output spec: +/-3% DC (4.85-5.15 V) over 18-30 Vin and 0-2 A load; ripple <= 50 mV peak-to-peak	Owner answered P0 Q3 with the default. Sets the regulation/ripple targets that bind feedback divider tolerance, output capacitance and the P8 sim bounds.

Phase	What	Why
P0	One switch-node probe pad plus an adjacent GND pad; no other test points	Owner answered P0 Q4 with the default. Mode-boundary ruling: the switch node is the block's own measurement need on a study board whose bring-up evidence proves the seeded records. Kept minimal and inside the existing SW polygon to limit added copper on the noisiest node.
P1	P1 research roster: research-power-architect + research-component-scout(buck-regulator) only. NO research-interface-spec, NO research-reference-design.	Screw terminals are bound by no electrical standard, so there is no interface spec to research. The buck block is not novel - the library already holds 16 buck records - and the useful vendor-layout digest is exactly what the P3 part-level coverage research produces for the CHOSEN IC; a P1 reference-design spawn would duplicate it. Mode says do not pad.
P1	Topology: synchronous integrated-FET buck, Fsw ~500 kHz (band 350-600), L = 10 uH	research-power-architect numbers: async costs +0.42 W (+33%) and adds a 0.77 W diode hot spot at Tj ~126 C against a 125 C rating, worst-cased by D=0.17 (diode conducts 83% of every cycle at 30 V). Fsw >= 1 MHz costs 1.58-1.93 W and collides t_on(30 V) with typical min-on-time. LDO is 38 W - arithmetic. Controller+FETs is the ladder's escape hatch for corners with no stocked integrated part; 30 V/2 A is not one.
P1	PROVISIONAL 2-layer stackup, conditional on the P3 part: sync + Rds_LS <= 85 mOhm + outline >= ~1000 mm2 + >= 16 thermal vias + Tj 150 C	The 2L screen passes by only 2 C (0.92 W x 73.8 C/W = 67.9 vs check_thermal dt_c 70). The layer count FOLLOWS THE PART - a 2 A-class part at Rds_LS 130 mOhm gives 1.19 W and forces 4L. P3 must re-run this arithmetic with the chosen part's real Rds before P5 fixes the stackup (no outline-shrink step exists; board_init is final).
P1	OPEN CONSTRAINT carried to P3 (not settled at P1): light-load ripple vs A3. A3 binds <= 50 mVpp across the FULL 0-2 A range INCLUDING no load, but PFM/burst light-load ripple is typically 50-100 mVpp at 5 V.	research-power-architect OPEN item. The mode excludes the usual fix (a preload resistor is excluded conditioning), so P3 part selection must supply ONE of: a forced-CCM part, a strappable MODE/FCCM pin, or a datasheet figure showing <= 50 mVpp at light load. If no JLC-stocked 36 V-class candidate offers any of the three, ESCALATE to the owner to choose between relaxing A3 at light load and admitting a preload against the mode. A burst-mode part must NOT ship silently against A3.
P2	SKIPPED the optional coverage-mapper spawn on the P2 mapping_request	Every one of the 15 buck records is ALREADY matched to block:B1 via keys and evaluated per class; the blockers are maturity-below-floor (8 classes), envelope-outside (inrush on source_kind, snubber on rectifier_kind, COT on control_kind) and level-below-min (1). A mapping edge cannot change any of those three verdicts, so a mapper pass is a provable no-op here. The only unkeyed library record (spiral-inductor-plane-and-heatsink-keepout) is also 'approved' and would still be maturity-blocked. Recipe step marks the mapper optional.

Phase	What	Why
P2	D5 OUTLINE 40 x 30 mm (1200 mm2) - FINAL, binds at P5 board_init	The outline IS the radiator: R_ba is 39/34/31 C/W at 875/1064/1200 mm2, so area buys more Tj than layers (~10 C vs ~11 C) and is free at the fab. 20% above P1's ~1000 mm2 floor. The increment over 38x28 is bought by physical parts - a 10-12.5 mm inductor (D11), two ~10 mm screw terminals, 4x M3 with 6.5 mm keepouts - not padding. No outline-shrink step exists later. J1 LEFT edge, J2 BOTTOM edge.
P2	D4 STACKUP JLC2313_1.6, 2 layers, 1 oz - CONDITIONAL	Conditions: sync + P_U1 <= 0.95 W at 30V/2A + outline >= 1000 mm2 with unbroken B.Cu pour + >= 16 thermal vias + Tj_max 150 C. Named escalation trigger: P_U1 > 0.95 W (at 400 kHz = Rds_LS > ~110 mOhm at 25 C; at 500 kHz, 85 mOhm) OR a 125 C part OR an async part OR check_thermal erroring at dt_c 70 -> 4L JLC04161H-1080B, outline unchanged. P3 MUST re-run this with the chosen part's real datasheet Rds BEFORE P5 freezes the stackup.
P2	D2 Lead part class: 36 V-class SYNC integrated-FET 3 A-class, exposed-pad SOIC-8-EP, fixed 400 kHz, internally compensated, peak-current-mode (LMR33630ADDAR). ADJUSTABLE output accepted.	CONFLICT RESOLVED: P1 preferred a FIXED 5 V part to delete the divider tolerance term; the scout's JLC sweep found no fixed 5 V sync 36 V-class part, and its only fixed candidate (XL1509-5.0E1) is +/-4 % guaranteed, which alone breaks A3. P1's PREFERENCE loses to stock reality - it was a preference, not a filter. Scout ranks 2-4 are asynchronous and are NOT drop-in fallbacks (D1 rejects async with numbers). If P3 finds a stocked fixed 5 V sync >= 36 V part, take it: R1/R2, the tolerance budget and the only sim candidate all disappear.
P2	D3 fsw 400 kHz, L = 15 uH (P1 recommended 500 kHz / 10 uH)	The lead part is a FIXED 400 kHz device with no RT pin, and 400 kHz sits inside P1's own 350-600 band. At 400 kHz L must be 15 uH (10 uH gives 52% ripple); P_U1 improves 0.924 -> 0.854 W. dI 0.694 A, I_L,pk 2.35 A, ripple 10.4 mVpp. Reverses if the part's internal-compensation L/C table excludes 15 uH - fallback 10 uH at 400 kHz (15.6 mVpp, I_L,pk 2.52 A, both in spec).
P2	D8 canonical net names +VIN /SW +5V GND (/FB and /BST deliberately NOT in constraints power[])	P1 proposed VIN/SW; renamed to match the KiCad export mechanism (power symbols export bare, root-sheet local labels export with one leading slash). A power[] entry on /FB would put a width rule on a high-impedance sense node and pull it into the check_pdn decoupling inventory. P4 must run netlist_audit.py - a rename surfaces as missing_net (error) and nothing else catches it.

Phase	What	Why
P2	D9 ONE FLAT ROOT SHEET, hierarchy rejected; D6 FB divider 0.1 %/25 ppm; D12 exactly ONE sim candidate (FB divider DC accuracy vs A3's window)	D9: 14 parts / 6 nets, and a child-sheet label exports as /<sheet>/<LABEL>, silently unhooking every constraints entry spelling it /<LABEL> (the class that cost lumina-par). D6: 1 % resistors give a worst-case SUM of ~3.7 % which fails A3; 0.5 % closes only on RSS; 0.1 %/25 ppm closes on the worst-case SUM (~2.5 %) for cents and no part-count change. D12: buck switching is not simulated by policy; the FB divider sweep catches the wrong-value class ERC/DRC/DFM are blind to, and disappears if P3 lands a fixed-output part.
P3	P3/P6/P7 spawns receive the knowledge prompt_block BY PATH (log/knowledge-P3.json, field prompt_block) instead of pasted inline	knowledge.py –select returned 24 records / 37067 chars. Pasting that inline into every P3/P6/P7 spawn would consume the orchestrator's context for content the orchestrator must never reason about anyway (rule 1: agents read the design, the orchestrator reads contracts). Files are the interface - the agent reads the same bytes either way, and the selection is sha-stable on disk. Deviation from the SKILL.md 'paste its prompt_block here' wording, recorded.
P3	D4 CONDITION MET - 2 LAYERS CONFIRMED. JLC2313_1.6 stands; no escalation to 4L.	Real datasheet Rds (LMR33630 DDA, p7 Electrical Characteristics): Rds_LS 66 mOhm typ / 110 mOhm MAX guaranteed across -40..+125 C (already worst-case, no extra hot-derating owed); Rds_HS 95/160. Conduction D*I^2*Rds_HS + (1-D)*I^2*Rds_LS at I_L,rms 2.01 A, D 0.167 = 286 mW typ / 478 mW worst-case; plus power_tree's switching+gate+Iq 286 mW = P_U1 0.57 W typ / 0.76 W worst-case against the 0.95 W gate. ~20% margin at worst case, versus the 2-7 C cushion the placeholder model predicted. The stackup is now free to freeze at P5.
P3	D3 CONDITION MET - L = 15 uH and a 2 x 22 uF C_OUT bank both sit inside the part's windows. No fallback to 10 uH needed.	15 uH clears the 3.5 uH slope-comp floor and sits at 23% ripple against the device max (inside the 20-40% band). Correction to the knowledge record's reading: the 52/72 uF C_OUT figure is TI's own LOAD-TRANSIENT sizing choice (datasheet 9.2.2.5 - 'usually limited by load transient requirements, rather than output voltage ripple'), not a stability floor. bb-buck carries no binding transient spec (power_tree s5 already calls the resulting sag normal), so the binding constraint is the ripple equation, already at 10.4 mV.
P3	D11 CONDITION MET - inductor Isat 8 A vs the 6.6 A floor, DCR 27 mOhm, body 12.3 x 12.3 x 8.0 mm	Floor is 1.3 x the part's max peak current limit I_SC 5.05 A = 6.6 A, not 1.3 x the 2.35 A operating peak. 21% margin. DCR 27 mOhm beats the 30 mOhm preferred target. Body sits at the TOP of the 10-12.5 mm class the 40 x 30 mm outline was sized around - it fits, with nothing to spare, which is what the outline decision anticipated.

Phase	What	Why
P3	RULED (orchestrator): J1/J2 screw terminal KF128-5.08-2P-AA at 250 V / 24 A ACCEPTED against the requirements' ASSUMED >= 300 V	The >= 300 V figure was an ASSUMED value the requirements-analyst wrote for a part that should never be the limiting element - not an owner requirement and not derived from the board. Actual board voltage never exceeds 30 V, so 250 V is 8x margin; current 24 A vs the ~2.5 A worst case. No single-piece 5.08 mm 2P THT SCREW terminal at >= 300 V is in stock - the 300 V options change the MECHANISM (two-piece pluggable, or spring-clamp), which is a worse trade than the voltage tier. Requirements section 2 assumption superseded.
P3	THREE part corrections vs architecture/blocks.md, all datasheet-cited; parts.json is the BOM of record for P4	(1) HF bypass C1 is 220 nF/50 V/X7R per datasheet 9.2.2.6, not blocks.md's 100 nF. (2) C4/C5 case is 1210, not 1206 - no 22 uF/25 V/X7R 1206 part exists with usable stock, and 1210 matches TI's own reference design. (3) **C_VCC 1 uF/16 V was MISSING from blocks.md entirely** - datasheet 9.2.2.8 calls it required for proper operation; added as C7. P4 builds from parts.json, so the omission cannot propagate, but blocks.md is stale on these three points and the design doc will show it.
P3	A3 AMENDED by owner: ripple <= 50 mV pk-pk holds from ~200 mA to 2 A (CCM region); larger PFM burst ripple accepted below ~200 mA. +/-3% DC accuracy is UNCHANGED and still holds at all loads including 0 A.	The LMR33630 DDA package has NO MODE/FPWM pin (8-pin table, none is mode select) and datasheet 8.4.1 states auto-mode only - PFM/diode-emulation at light load, explicitly 'yields larger output voltage ripple'. The only ripple evidence is Fig 8-7, a TYPICAL unguaranteed capture at Vin 12 V / Iout 10 mA, which is neither 0 A nor this board's 18-30 V range: none of the three accept criteria (forced-CCM / MODE pin / datasheet figure at this operating point) is satisfied. Owner ruled to relax rather than override the mode with a preload or re-source. PFM at light load is a deliberate efficiency feature of nearly every modern buck, and on a study article the burst behaviour is itself instructive. P8 sim bounds take the 200 mA threshold.
P3	Min on-time for THIS package is 75-108 ns (DDA), NOT the 68 ns headline - and it is not binding	datasheet_extract: the Features-bullet 68 ns is the RNX/VQFN value; Table 7.6 gives DDA-specific 75 ns typ / 108 ns max. Required t_on at the 30 V corner is 5/30/400 kHz = 417 ns, so even the 108 ns max leaves ~3.9x margin. The P1 selection filter (t_on_min <= 130 ns) is satisfied by the correct number, not only by the optimistic one.
P3	Thermal vias under U1: >= 16 (P1/D4 thermal condition), which also clears the datasheet's own 4x3 = 12 minimum. Figure 10-2's ~6 vias is NOT the spec.	datasheet_extract found a genuine intra-datasheet discrepancy: section 10.1.1 text mandates a minimum 4x3 array of 10-mil thermal vias (12) while the worked example Figure 10-2 draws only ~6. The extraction records 12 as binding text and flags the mismatch so P6/P7 cannot silently copy the smaller figure count. Our own 2L thermal case was computed at >= 16, which is the stricter number and therefore governs.

Phase	What	Why
P3	FOOTPRINT RISK for the librarian/fp_verify: the datasheet carries TWO DDA mechanical revisions with DIFFERENT exposed-pad sizes - DDA0008J (03/2016) and DDA0008B (09/2025). Extraction used the NEWER DDA0008B (pad copper 2.71 x 3.4 mm).	A pulled or stock footprint built to the 2016 drawing will have the wrong exposed-pad size against the extraction. fp_verify must be checked against DDA0008B's 1.27 mm pitch, 8 x (0.6 x 1.55 mm) pads, 5.4 mm row spacing and 2.71 x 3.4 mm pad copper. Also note the pinout table itself does not number the pad - 'pin 9' comes only from the mechanical drawing's corner callouts, which is exactly the kind of mismatch that breaks a symbol-to-footprint pin map.
P3	OPEN RISK carried to P4 review: C_OUT is 2x22uF/25V where TI's own DDA/400 kHz/5 V reference BOM uses 4x22uF/16V. Confirm 2x is inside the part's STABILITY window, not merely its ripple budget.	The part-sourcer justified 2x by finding the 52/72 uF figure in record buck-cmode-internal-comp-cout-window is TI's LOAD-TRANSIENT sizing choice (9.2.2.5), not a stability floor, and bb-buck has no binding transient spec - sound for transient. But the extraction independently reports TI's reference BOM as 4x22uF/16V, and an internally-compensated current-mode loop places its output pole from C_OUT: 2x22 uF at 25 V derates under 5 V bias to ˜32 uF effective vs ˜66 uF for TI's 4x. Narrow question for the P4 schematic-reviewer to answer FROM THE DATASHEET: is a minimum C_OUT stated for loop stability (as distinct from transient), and does ˜32 uF effective sit inside it? If not, the fix is cheap now (add C8/C9, same MPN - the 40x30 outline has room) and expensive after P6.
P3	RULED (orchestrator): ACCEPT the vendor U1 footprint as pulled (signal pads 0.6 x 1.3 mm vs DDA0008B nominal 0.6 x 1.55; EP 2.5 x 3.5 vs 2.71 x 3.4). No redraw.	fp_verify rates it a WARNING, not an error, and pitch 1.27 mm and row spacing 5.4 mm are both exact - the geometry that determines whether the part lands correctly is right. The deviation is small and in the safe direction: shorter toes and a slightly narrower EP mean MORE clearance, not bridging risk; EP area 8.75 vs 9.21 mm2 is a 5% thermal delta against a case that already carries ˜20% margin. Decisive argument: this is the LCSC/EasyEDA footprint that JLC's own assembly process is built around, so redrawing to datasheet nominal trades a paper match for a mismatch with the house that actually builds the board. Cost of being wrong is a slightly smaller toe fillet on a bench article, and T8's swap_new_fp makes a later change non-destructive. Owner can override.

Phase	What	Why
P3	RULED (orchestrator): ACCEPT the courtyard-cuts-through-pads defect on C0603/R0603/C1210/L1. No redraw.	Documented library-wide easyeda2kicad class (courtyard = body outline only, smaller than the pad span). It is NOT silently breaking placement legality: since the T-fix, pairwise overlap tests the PRECISE shape - declared courtyard UNION pad bbox - so placelib already accounts for the pads the courtyard omits. All 6 footprints HAVE an F.CrtYd (none courtyard-less, which is the case that genuinely degrades the place gate). No clean redraw convention exists yet beyond '0.25 mm beyond pad-field bbox'. Revisit as a library-level fix, not per-board.
P3	No new research opened at the P3 exit despite block:B1 re-firing at the proven floor	The mode's first-call escalation (–maturity-floor proven –research-provisional) re-reports block:B1 as gap because no buck record is yet PROVEN and only bring-up evidence can retire that - it is the same slot already researched at P2 (task block-buck-1, 9 records, 9 verified). Re-opening it would re-research work just done and burn a second per_run slot for nothing; the mode's own instruction is to re-run WITHOUT the flags after research so the re-run folds in what was learned. The genuinely NEW P3 slot - the part slot (datasheet-layout) - reads COVERED off the C841384 extraction's 16 layout bullets, which is exactly the P3-exit test. Plain re-run: 0 gaps.
P4	ONE schematic agent for the whole board (recipe's small-board amendment), not one per sheet	P2 decision D9 rejected hierarchy: 14 parts / 6 nets on ONE FLAT ROOT SHEET. There is exactly one sheet, so a per-sheet split would spawn one agent and a stitcher for the same file. Recorded per the full-run recipe: 'Small boards may use ONE schematic agent (record it).'
P4	APPROVED after the fact: schematic agent retyped lib/aiee.kicad_sym U1 pin 7 (BOOT) from power_in to passive	Evidence-first, and the agent showed its work: with the retype disabled the finished sheet produced EXACTLY ONE ERC finding, power_pin_not_driven on U1 pin 7, and nothing else. /BST has no schematic-visible driver because the boot diode is inside the package. The PWR_FLAG alternative would clear ERC but drag /BST into netlist_audit power_undeclared, and sheets.md s4 deliberately leaves /BST undeclared. Precedent: sbuck-5v3a's AP64350 BST pin carries the same passive typing. Deviations are listed in gen/lib_pin_types.py so a later lib_pull cannot silently re-type wholesale.

Phase	What	Why
P4	E1 CONFIRMED (reviewer, error): C_OUT 2 x 22 uF is OUTSIDE the datasheet's stability-scoped range. FIX = add C8/C9, same MPN C2918511, to reach 4 x 22 uF.	The earlier phase was right that Eq 6 is a transient bound, but never reached Sec 9.2 p19: 'the internal compensation is optimized for a certain range of external inductance and output capacitance' - quantified ONLY in Table 9-2 p20, whose 400 kHz / 5 V row this board matches on EVERY entry (RFBT 100k, RFBB 24.9k, CBOOT, CVCC, CFF open) except L and C_OUT, and which specifies 4 x 22 uF RATED. Every 400 kHz row is 4x; 2x appears only at 1400/2100 kHz. TI's own DDA characterization BOM (Table 9-3 p31) used 4 x 22 uF at this exact operating point. Physics: fixed internal Type-II compensation makes fc ~ 1/C_OUT exactly, so halving C_OUT roughly doubles crossover toward fSW/2 = 200 kHz, on an all-ceramic bank whose ESR zero sits near 3 MHz and contributes no phase boost. The failure is SILENT at DC and appears only on load/line steps - which is precisely why no gate on this board would have caught it. Derating does not rescue 2x: effective ratio to TI's 16 V bank (~0.52) tracks the rated ratio (0.50). Cost ~$0.45/board, free now, expensive after P5/P6.
P4	W1 will be FIXED not waived: input bank 2 x 10 uF / 50 V straddles the datasheet's 10 uF EFFECTIVE minimum. Add a third 10 uF (C10, same MPN C2918502).	Sec 9.2.2.6 requires 10 uF MINIMUM and Sec 9.1 p19 defines that as EFFECTIVE, not nameplate. 2 x 10 uF / 50 V gives 10-12 uF effective at 30 V bias, falling to 8.5-10.5 uF with tolerance and the 85-90 C board temperature constraints.json predicts - i.e. BELOW the minimum in the worst corner. It is only a warning because it straddles rather than clearly fails, but the schematic is being regenerated for E1 anyway, the part is ~$0.08, and the 40 x 30 mm outline has room. Fixing is strictly better than waiving when the regeneration is already happening. NOTE the mode does NOT argue against this: ultra-bare-bones explicitly INCLUDES the support components the datasheet requires at the stated operating point, so under-populating was the mode violation, not over-populating.
P4	W2 is a LAYOUT carry-forward to P6/P7, not a schematic fix: R2's ground return must be pinned to the AGND/EP reference	Reviewer: nothing currently enforces it - constraints.json's feedback group covers only /FB routing, and the _review_enforced item 5 covers only Cin/Cout returns. The exposed pad IS AGND (Table 6-1: all electrical parameters are measured with respect to it), so the divider's bottom-leg return is a reference-accuracy path, not just a ground connection. Carry into the placement and routing spawns explicitly.

Phase	What	Why
P4	CORRECTION to the earlier A3 amendment: DC accuracy at 0 A is NOT unaffected by PFM. Owner ruled BOTH remedies: re-centre the divider to 102k/25.5k AND extend the light-load carve-out to cover DC accuracy below ˜200 mA.	sim-analyst finding, from datasheet SNVSAN3F Sec 7.7: VOUT regulation at 5 V is -1.5/+1.5 pct for IOUT >= 1 A but -1.5/**+2.5 pct** for IOUT 0 A to max - a +1.0 pct light-load adder, stated outright in Sec 8.4.1 (PFM / frequency foldback raises Vout at light load) and visible as a 5.01-5.06 V band in Fig 9-19. A3 explicitly requires 0 A. Stacked on the bench's binding 0-50 C corner: 5.10958 x 1.010 = 5.161 V, MISSING the 5.15 V ceiling by 11 mV. Root cause is NOT the resistors (they own only 0.26 pct of the +2.19 pct budget; VREF and the setpoint offset own the rest) but the +0.32 pct setpoint offset - TI's own Table 9-2 pair sets 5.0161 V, not 5.000. My earlier statement that DC accuracy was unchanged at all loads was wrong; same PFM root cause as the ripple carve-out already accepted.
P4	R1 100k -> 102k (C861068) and R2 24.9k -> 25.5k (C861257): exact 4.000 ratio, 5.000 V nominal. Both E96, both 0.1 pct / 25 ppm, same Yageo RT family (TCR tracking preserved), both verified in stock (2009 / 18845 vs 25 needed).	Recentring moves the stacked worst case from 5.161 V (outside) to ˜5.144 V (inside by 6 mV) and costs nothing - same package, same family, same price class, no library work (R0603 already pulled). It deliberately departs from TI Table 9-2's 100k/24.9k row, which is acceptable because the divider is a SETPOINT item, not a stability item: the ratio sets Vout, and the absolute values matter only for the 1 M CFF threshold (102k is far below it) and for noise, where 102k is the same impedance class.
P5	Outline strategy: board_init at 35x25 with 2 diagonal M3 holes; MEASURE the placed extent at P6 and, if there is real slack, rerun board_init at the measured size and RE-PLACE before routing.	Owner wants a tight board sized by the layout rather than chosen up front. Verified (not just cited): –outline exists ONLY on board_init.py and no other script writes Edge.Cuts, so resizing rebuilds the board from the netlist and discards placement AND routing; hand-editing the .kicad_pcb is barred by rule 1 and by the board-setup contract. Placement is cheap to redo and routing is not, so the correct order is place -> measure -> resize+replace if warranted -> route ONCE at the final size. 4 holes -> 2 diagonal frees ˜85 mm2 of corner keepout on an 875 mm2 board; requirements.md s5 explicitly sanctions two on opposite corners when four stop being honest.

Phase	What	Why
P6	SCRAP AVERTED: both provisional connector rotations were WRONG and are corrected - J1 0 -> 270, J2 90 -> 0	P2 flagged both rot values PROVISIONAL because the footprint's native wire-entry direction could not be verified before a footprint existed. It could not: the WRL carries (rotate 0 0 180), the documented coincidence trap. The placement agent resolved the true native entry (+y at rot 0) on a 3D iso test board and then CONFIRMED the finished board with orthographic left/front renders - J1's mouths face -x off-board, J2's face +y off-board. As declared, BOTH openings would have pointed inward: the terminals would be unusable and the boards scrap. This is exactly why the role prompt forbids taking seed rotation on faith.
P6	Annealer candidates ALL REJECTED on structure; the hand placement is the winner (it also won on hpwl 125.0 vs 130.3-130.8)	Every candidate put the FB divider hard against L1's SW pad and TP1/TP2 outside the switch-node copper - precisely what the cost function cannot see (buck-fb-route-rules). Structure beats score. The hot loop was hand-placed and LOCKED before anneal per the records (the annealer cannot discover a hot loop): C1 220 nF sits 1.72 mm from BOTH VIN(2) and PGND(1) with pads UNCROSSED, ˜2.2 mm2 loop. A rotation-sign error that had mirrored C1/C6 (+VIN facing PGND, forcing the loop to cross itself) was caught and fixed mid-run.
P6	OUTLINE IS FINAL AT 35 x 25 - measured, not estimated. Do NOT resize; route here.	Occupied extent is 34.90 x 24.90 mm of courtyards with 0.05 mm slack on ALL FOUR edges. Fill 72.7 % (636.1 / 875 mm2); the 24.6 % free area is one connected web of interstitial channels whose largest inscribed empty rectangle is 4 x 4 mm - ONE more 1210 laid flat does not fit anywhere on the board. The all-round 0.05 mm is set by H1/H2's 6.9 mm courtyards at their constraint-fixed centres, so any shrink drags both washer keepouts onto the hot loop and C4. Ignoring the holes the true recoverable slack is ˜1.2 mm width / ˜0.6 mm height = at best 34 x 24 (-7 %), and spending it would consume the 0.15-0.40 mm inter-part gaps this layout already runs on. The owner's place-measure-resize plan is therefore ANSWERED: no resize, and routing is not put at risk.
P6	RULED (orchestrator): ACCEPT L1's footprint aiee:IND-SMD_L12.5-W12.5_RLF12545T despite the name referencing a different vendor's part (TDK RLF12545T) than the sourced SMDRI127-150MT	Same reasoning that settled the U1 land, and it is the decisive one: lib_pull keys on the LCSC CODE, not on a name. The footprint was pulled for C40000, so it IS the land pattern LCSC/JLC associates with the part being ordered - EasyEDA simply reused a generic 12.5 mm-class inductor land-pattern NAME. Redrawing to some other vendor's drawing would trade a name match for a mismatch with the house that assembles the board. Note the 3D render's '470' marking is likewise a WRL artifact of the donor model, not our 15 uH part - cosmetic only, no electrical or fab meaning. Librarian OPEN item 4 closed.

Phase	What	Why
P7	APPROVED: router REVERTED route_critical –only power and routed the power nets deliberately instead	KRT put /SW (2 seg + 3 vias), +VIN (1 seg + 2 vias) and +5V (6 seg + 3 vias) on B.Cu, slicing the ground shield through the hot loop and the switch node - a direct hit on the binding constraint that the unbroken bottom pour is the ONLY thing recovering this 2-layer board's ˜5 dBuV/m deviation from the datasheet's own 4-layer layout rule. Its +5V path also swung around the right edge past H2. Reverting a tool that is optimising the wrong objective is the correct call, not a workaround. Result: 36 tracks, ALL F.Cu, ZERO B.Cu tracks board-wide, zero vias on +VIN / /SW / +5V.
P7	ACCEPTED DEVIATION: route_auto (Freerouting) was NOT run	After the deliberate route the board already measured DRC 0 / completion 1.00 (33/33 ratsnest, 0 unconnected_items). Exporting a DSN of a fully-coppered board is the documented Freerouting wedge case (LEARNINGS 435), and its KRT finish can ADD B.Cu copper - which is exactly the failure just reverted. The role prompt is explicit that Freerouting's own success signal is untrusted and only kicad-cli DRC gates; completion here is measured from DRC, the authoritative source. Running it could only risk the shield for no measurable gain.
P7	ACCEPTED: +5V reaches the left output-cap bank (C5/C8/C9 + R1) through an 18.3 mm2 pour bridge with a 1.00 mm neck, not a 1.52 mm track	Sanctioned track_width remediation step 4 (pour, do not neck): a 1.52 mm track geometrically cannot reach that bank because placement split it and the R1.2/TP2/C5.2 pinch caps any track at ˜0.95 mm. The DC LOAD path is untouched - L1.2 -> J2.1 at 1.8 mm - so only capacitor ripple crosses the bridge, where at 400 kHz the neck's few nH is milliohms against the caps' own impedance and cannot undo the E1 stability fix. FB senses at C5.1 on the far side of the bridge, which carries no DC, so the sense point sees no bridge drop; the unsensed L1-to-J2 drop is ˜2.7 mOhm = ˜5.5 mV at 2 A against a 150 mV window.
P8	WO-2 OVERRIDDEN: the two check_pdn 'no decoupling capacitors' errors (+5V, GND) go to verify-waivers.json, NOT to a schematic fixer	fix_dispatch routed them to the schematic domain, which would be wrong work: you CANNOT decouple a return net, and +5V is a regulator OUTPUT whose decoupling IS the C4/C5/C8/C9 bank - those caps sit on no IC power pin, so they are never registered as decoupling associations and the check cannot see them. P2 decision D7 pre-ruled this exact finding a category error destined for waivers, before the board existed. Adding capacitors to satisfy it would be mode-excluded parts solving a measurement artifact. WO-1 (silk) and WO-3 (pour neck) are real and proceed.

Phase	What	Why
P8	check_current F.Cu pour neck resolved by a constraints overrides region (rung 4), NOT a waiver - and check_irdrop independently corroborates it	The fixer proved no copper fix exists: pouring GND at the absolute 0.127 mm DRC floor over every free mm gives a 0.523 mm ceiling and deleting EVERY foreign F.Cu track still only reaches 1.134 mm against 1.520 required; it passes only by deleting capacitor pads, i.e. moving the hot loop. It also proved the neck is not in series with the rail (F.Cu splits into 5 leaf-tap via groups each with its own 3-via drop to B.Cu; both GND terminals are THT so the return enters the bottom plane directly). INDEPENDENT CONFIRMATION from a checker that knows nothing about pour necks: check_irdrop's 2.5D solve gives GND a total resistance of 0.233 mOhm and a 0.61 mV drop at 2.6 A - TWENTY TIMES LOWER than +5V's 4.855 mOhm / 12.68 mV. A 0.32 mm series neck would have made GND the worst net on the board; it is the best by an order of magnitude.
P8	FIXED A CHECKER, NOT THE BOARD: check_silk built every circle as a filled disc regardless of (fill no), so stock KiCad ring footprints reported their own pads as 100 pct silk-covered	check_silk.py line ~172 returned Point(center).buffer(r + w2) unconditionally. The stock TestPoint_Pad_D1.5mm silk ring is r 0.95 with a 0.12 stroke = an annulus spanning r 0.89-1.01 mm, against a pad of r 0.75 - i.e. 0.14 mm of CLEARANCE, no contact at all. The reported overlap of 1.77 mm2 is exactly pi*0.75^2, the whole pad, which is the signature of the disc assumption. Waiving it would have recorded a non-defect as an accepted defect on a board whose probe pads the owner will actually scope. Fixed with an _is_filled() helper returning an annulus for unfilled circles. rect/poly remain solid-buffered - conservative (over-reports) rather than unsafe, and noted as the same latent class.
P8	P8 layout legs both PASS: check_irdrop 0 violations (grid converged, richardson_delta 0.0015 on +VIN), check_pdn_z 0 violations	+VIN 6.97 mV at 1.1 A; +5V 12.68 mV at 2.6 A (against a 150 mV A3 window); GND 0.61 mV. /SW correctly skipped (pdn:false, width-only entry). check_pdn_z skipped +VIN and /VCC for want of an adjacent-layer fill pair, which is simply what a 2-layer board is - not a coverage gap.

Phase	What	Why
P8	CORRECTION to my own P3 decision: '>= 16 vias within 4.6 mm' does NOT clear the datasheet's 4x3=12 minimum - those count DIFFERENT things, and the board is short in-pad (9 vs 12)	My P3 decision recorded that our >= 16 requirement 'also clears the datasheet's own 4x3 = 12 minimum'. That conflated two different populations. The datasheet Sec 10.1.1 minimum is 12 vias UNDER THE PAD; our >= 16 counts vias anywhere within 4.6 mm, which includes the surrounding F.Cu island ring - a far longer heat path with much lower barrel conductance. The board has 18 within 4.6 mm but only **9 in-pad** (3x3 at 0.85 mm pitch), i.e. ~88 pct of the datasheet's barrel-conductance floor, worth ~5 K of Tj - most of check_thermal's 2.06 C reported margin on the sole heat path of a 2-layer thermal design. Caught by the fresh-context verify-reviewer, which is exactly what it is for. A 3x4 array fits with 0.3 mm to every pad edge.
P8	EP in-pad vias 9 -> 12 (3x4 at 0.85 mm), barrel conductance 87.4 -> 116.5 pct of the datasheet floor, -4.91 K Tj. Gates unchanged PASS.	The array was RE-SPACED, not appended to: a 4th row on the existing 3x3 would need y = +/-1.7 against a pad half-height of 1.75, i.e. outside the pad. 12 vias at 0.30 mm now beat the datasheet's 12 x 10-mil floor on conductance as well as count. All 12 annuli inside pad copper; hole-to-hole 0.55 (floor 0.5), annulus 0.15 (floor 0.15); zero B.Cu tracks created; AGND spine untouched.
P8	GATE BLIND SPOT, measured: check_thermal CANNOT see the thermal-via fix at all - margin 2.0623 C before and 2.0623 C after	theta_ja() is a function of copper AREA and LAYER COUNT only; vias_near_part is computed and reported but never enters the model (check_thermal.py:55-56, 93, 131). So on a 2-layer board whose entire heat path is the EP via array, the thermal gate is insensitive to the array. This is exactly what let the 9-vs-12 shortfall through P3 unnoticed - and it means the 2.06 C 'margin' the gate reports is not a measurement of this board's real thermal case. Script proposal recorded: derive an in-pad via count from the part JSON's exposed_pad.thermal_vias (present here, value 12) as a first-class check, so the class is caught by a script rather than by a fresh-context reviewer.

Phase	What	Why
P8	Terminal silk: 4 of 6 labels placed (J1 '+' and 'VIN'; J2 '-' and '5V OUT'). The requirement is MET despite the shortfall.	On a TWO-PIN terminal a single polarity glyph fully determines the pinout: J1's '+' on pin 1 makes pin 2 negative by construction, and J2's '-' on pin 2 makes pin 1 positive. The two unplaced marks were redundant, not load-bearing. The fixer ran an exhaustive whole-board grid search (0.02-0.1 mm steps, all rotations, real silk/pad/track/courtyard obstacles, cross-checked against live DRC on a scratch copy) and found ZERO positions for them that were simultaneously legible, non-overlapping and closer to their own pin than to the sibling pin - the C5/C8/C9 bank hems in both connectors on exactly that side. It declined rather than place a misattributing label, which is the correct call ('a label readable against the wrong pin is worse than none'). RESIDUAL, for the owner at H4: J1's '+' sits above the connector body and left of C10, so a reader could conceivably attribute it to C10 - mitigated by 'VIN' sitting directly above it, but not eliminated.

Sheet plan

sheets.md - bb-buck schematic sheet plan, refdes ranges, CANONICAL net names

1. Sheet plan: ONE FLAT ROOT SHEET

```
| Sheet | File | Blocks it carries | Interface nets at its boundary | Refdes
ranges | ‘pwr_base‘ |
|---|---|---|---|---|---|
| root (only) | ‘kicad/bb-buck.kicad_sch‘ | B1 buck converter, J1 input, J2
output, TP1/TP2 probe, H1-H4 mounting | none - the board’s only interfaces are
the two physical terminals (‘+VIN‘/‘GND‘ at J1, ‘+5V‘/‘GND‘ at J2) | ‘U1-U9‘,
‘L1-L9‘, ‘C1-C19‘, ‘R1-R9‘, ‘J1-J9‘, ‘TP1-TP9‘, ‘H1-H9‘ | **1** (‘#PWR01..‘,
‘#FLG01..‘) |
```

A hierarchical split is deliberately REJECTED on this board. Hierarchy exists to keep large schematics readable and to give each sheet its own refdes/#PWR range; this board has 14 electrical parts and 6 nets on one A4 page. The cost of splitting is concrete and has been paid before: a label inside a child sheet exports as `/<sheet>/<LABEL>`, which silently unhooks every `constraints.json` entry that spells it `/<LABEL>` (the P4 amendment class that cost `lumina-par`, and the reason `sbuck-5v3a` also went flat). With one root sheet there are no sheet prefixes anywhere, by construction.

Refdes ranges are therefore trivially unique. They are stated anyway because the rule is "unique ACROSS sheets": **if a later amendment ever splits this sheet, the child sheets take `pwr_base = 40` and `80` and keep the prefix ranges above**, and every net name in s2 must be re-checked against the new export.

2. CANONICAL NET NAMES - these are now binding

`research/power.json` proposed `VIN / SW / +5V / GND`. **P2 makes them canonical here, and two of the four change spelling.** These are the names the schematic MUST produce and the names `constraints.json` uses.

Canonical net	KiCad mechanism	Spans	Declared in constraints.json as
‘+VIN‘	power symbol (VALUE = net name) -> exports BARE	J1.1 -> C1/C2/C3 -> U1 VIN pin	‘power‘ 1.1 A, ‘voltages‘ 30 V
‘GND‘	power symbol -> exports BARE	everything; B.Cu pour	‘power‘ 2.6 A ‘plane_fed‘, ‘planes‘ F.Cu + B.Cu, ‘thermal‘ heatsink net
‘+5V‘	power symbol -> exports BARE	L1 -> C4/C5 -> R1 -> J2.1	‘power‘ 2.6 A
‘/SW‘	root-sheet LOCAL LABEL -> exports with ONE leading slash	U1 SW pin -> L1 -> TP1	‘power‘ 2.6 A ‘pdn:false‘, ‘high_speed‘, ‘voltages‘ 30 V
‘/FB‘	root-sheet local label	R1/R2 midpoint -> U1 FB pin	**deliberately NOT declared** (see s4)
‘/BST‘	root-sheet local label	U1 BOOT pin -> C6 -> ‘/SW‘	**deliberately NOT declared** (see s4)

Why `+VIN` and not `VIN`: it is a rail feeding a power pin and decoupled by the input bank, so it is a power symbol, and a power symbol's exported name is bare. Why `/SW` and not `SW`: it is a root-sheet local label, and KiCad prefixes those with a slash on export. **A power symbol WINS over a coincident label**, so P4 must not mix the two mechanisms on one node.

Enforcement, not hope: at P4 run `netlist_audit.py --net kicad/bb-buck.net --constraints architecture/constraints.json`. A net spelled differently in the schematic surfaces as `missing_net` (error). Without that run, a misspelling is silent: `check_current` finds no copper on a net that does not exist and reports nothing.

3. Contents of the root sheet

Refdes	Part	Net attachments
‘U1‘	36 V-class synchronous integrated-FET buck, 3 A-class, exposed pad, 400 kHz, internally compensated	VIN=‘+VIN‘, GND/PGND/EP=‘GND‘, SW=‘/SW‘, BOOT=‘/BST‘, FB=‘/FB‘, EN per datasheet (see below)
‘C1‘	100 nF 50 V X7R - HF input bypass, **‘role: reg_input‘**	‘+VIN‘ / ‘GND‘
‘C2‘ ‘C3‘	10 uF 1210 50 V X7R - input bank	‘+VIN‘ / ‘GND‘
‘C6‘	100 nF - bootstrap (value/rating per datasheet)	‘/BST‘ / ‘/SW‘
‘L1‘	15 uH shielded/composite, <= 40 mOhm, Isat >= ~6.6 A	‘/SW‘ / ‘+5V‘
‘C4‘ ‘C5‘	22 uF 1206 25 V X7R - output bank	‘+5V‘ / ‘GND‘
‘R1‘ ‘R2‘	FB divider, 0.1 % / 25 ppm (‘R1‘ = top, ‘+5V‘->‘/FB‘; ‘R2‘ = bottom, ‘/FB‘->‘GND‘)	as listed
‘J1‘	2-pole 5.08 mm screw terminal, THT	1=‘+VIN‘, 2=‘GND‘
‘J2‘	same part as J1	1=‘+5V‘, 2=‘GND‘
‘TP1‘	SW probe pad (A4) - INSIDE the SW copper, not on a spur	‘/SW‘
‘TP2‘	GND probe pad, adjacent to TP1	‘GND‘
‘H1‘-‘H4‘	M3 clearance holes, 3.2 mm	mechanical (no net)

EN: read the part's own behaviour before adding anything. Most parts in this class auto-start from an internal pull-up, so EN ties to `+VIN` or floats and NO parts are added. No UVLO divider - mode-excluded, and a bench supply has no cable-droop motorboating case (0.62 A over 2 m of lead is ~0.1 V). No soft-start components: internal soft-start makes Cout inrush `C x Vout/tss` ~ 65 mA, and A2 means the supply ramps from zero anyway.

If the FB pin has no divider because P3 found a FIXED 5 V part, `R1/R2` and `/FB` disappear and the sense point ties straight to the output - **that is the preferred outcome and it removes the whole s5 tolerance budget**, but do not synthesise it from an adjustable part's datasheet figure.

4. Two nets deliberately left OUT of `constraints.json`

`/FB` and `/BST` are declared nowhere. This is intentional:

- `rules_gen` buckets every `power` entry into a netclass **at that net's own IPC-2152 width** (this used to flatten every power net to the widest width; the current script buckets per net - verified in `rules_gen.net_classes`). Even so, a `power` entry on `/FB` would give the feedback node a minimum-width rule and pull it into the `check_pdn` decoupling inventory, and `/FB` is a high-impedance sense node that wants to be SHORT and THIN and nowhere near `/SW`.
- `/BST` carries only gate charge. Nothing to size, nothing to decouple.

The two rules that DO apply to them are layout rules, carried in s5 below and in `constraints.json` `placement.groups` / `separation`, not width rules.

5. Placement groups this sheet implies (they become P6 groups)

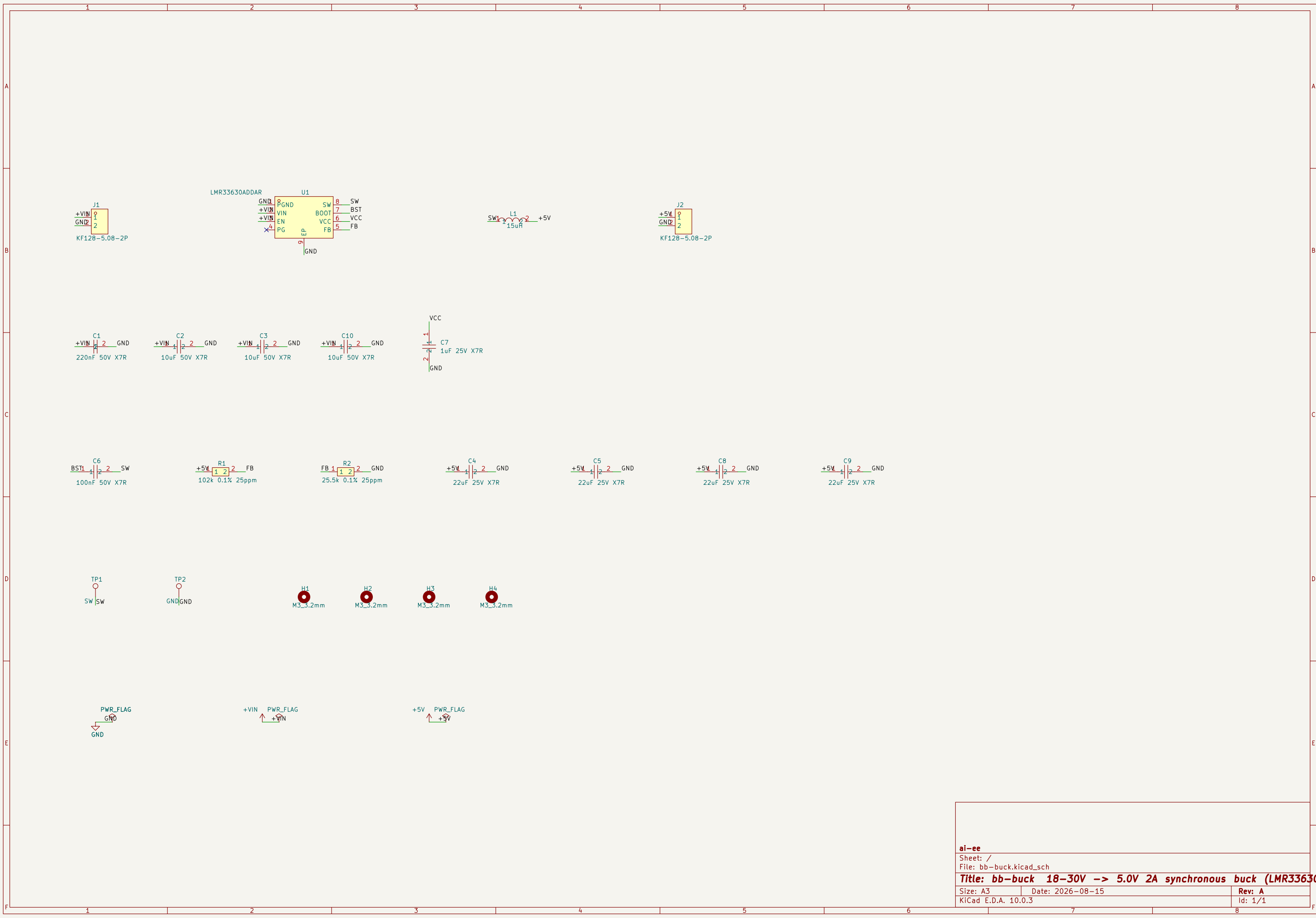
- **'hotloop'** (anchor `U1`; `C1`, `C2`, `C3`, `C6`): C1 nearest the VIN pin, then the bulk ceramics, then U1 VIN -> U1 PGND, on F.Cu, **smallest enclosed area achievable**. This is the loop that spends ~37 mV of the 50 mV ripple budget. It must be placed by explicit `place_edit` and LOCKED before `place_anneal` runs - an annealer optimises a cost function and cannot discover a hot loop.
- **'output'** (anchor `L1`; `C4`, `C5`): Cout ground and Cin ground return to the GND pour at SEPARATE points near their own pins - never a shared narrow neck (`buck-cin-co-ground-separation`).
- **'feedback'** (anchor `R1`; `R2`): at the FB pin, sense point AFTER the output caps, >= 5 mm from L1 and off the `/SW` side of the package.
- **'probe'** (anchor `TP1`; `TP2`): TP1 inside the existing `/SW` pour; TP2's ground returns straight down to the B.Cu pour beneath it.

`/SW` itself: a short WIDE pour, 1.6-2.0 mm wide, <= 8 mm long, <= 40 mm^2 total, F.Cu only, never near a board edge or a screw terminal. IPC-2152 constrains the WIDTH (a floor); EMI constrains the AREA (a ceiling that nothing in the toolchain enforces). TP1's pad counts against that 40 mm^2.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

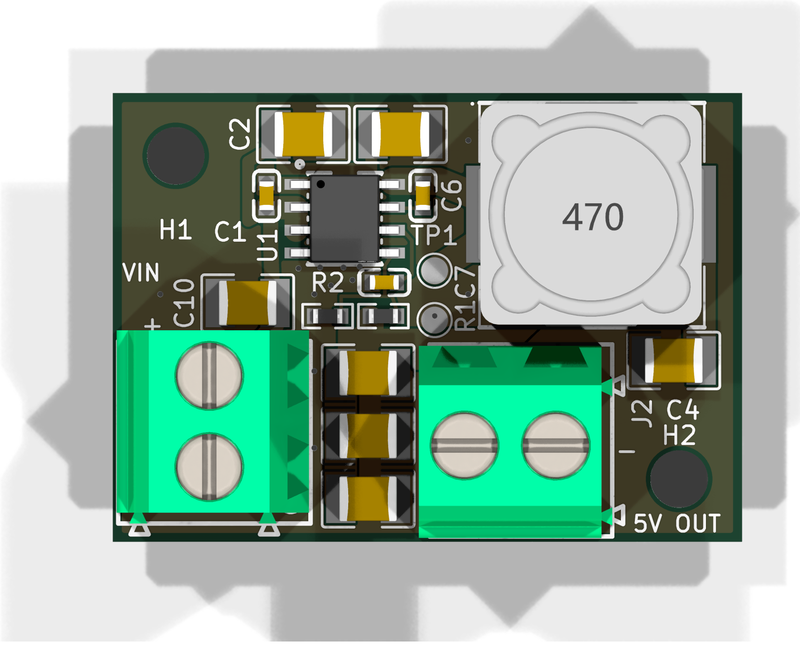
Gate `erc`: *not recorded.* The full schematic PDF follows.



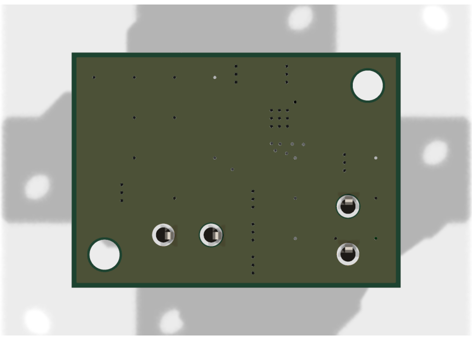
LMR33630ADDAR
U1
GND
VIN
EN
PG
SW
BOOT
VCC
FB
BST
J1
KF128-5.08-2P
J2
L1
15uH
+5V
+VIN
C1
220nF 50V X7R
C2
10uF 50V X7R
C3
10uF 50V X7R
C10
10uF 50V X7R
C7
1uF 25V X7R
C6
100nF 50V X7R
R1
102k 0.1% 25ppm
R2
25.5k 0.1% 25ppm
C4
22uF 25V X7R
C5
22uF 25V X7R
C8
22uF 25V X7R
C9
22uF 25V X7R
TP1
TP2
H1
H2
H3
H4
M3_3.2mm
PWR_FLAG
ai-ee
Sheet: /
File: bb-buck.kicad_sch
Title: bb-buck 18-30V -> 5.0V 2A synchronous buck (LMR33630A)
Size: A3
Date: 2026-08-15
Rev: A
KiCad E.D.A. 10.0.3
Id: 1/1

6 Layout

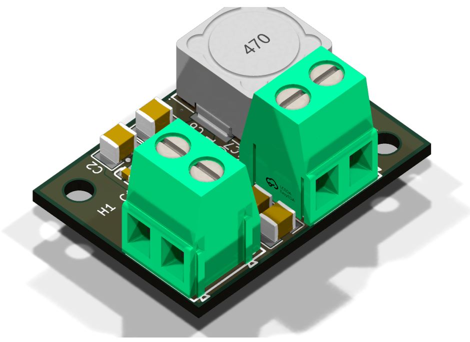
Gate `place`: *not recorded.*
Gate `drc_routed`: *not recorded.*



reports/renders/bb-buck_top.png



reports/renders/bb-buck_bottom.png



reports/renders/bb-buck_iso.png

7 Verification

Gate `verify`: *not recorded. reports/verify_all.json not available.*

Design review of record

P8 board review - bb-buck (adversarial, fresh context)

2 errors / 11 warnings / 3 waivers recommended. Machine checks were green when I started; everything below is what they cannot see, plus a verdict on every warning they did emit and an audit of the three resolutions taken this phase.

Renders I generated and reference below (all in `reports/renders/`): `bb-buck_top.png`, `bb-buck_bottom.png`, `bb-buck_iso.png`, and two elevation views I added specifically to settle the terminal question: `bb-buck_edge-left.png` (camera on the LEFT board edge) and `bb-buck_edge-bottom.png` (camera on the BOTTOM board edge).

Coordinates are board-global mm (the same frame the check reports use); board-local mm are given in brackets where it helps. Board origin offset is (26.07, 34.48).

—

ERRORS

1. Neither screw terminal carries any polarity or function silk (silk) `J1` (31.670, 53.380) and `J2` (48.770, 53.880).

Every one of the 20 F.SilkS text items on this board is a refdes. There is no `+`, no `-`, no `VIN`, no `5V`, no `OUT` anywhere on F.SilkS or B.SilkS, and there are zero board-level text objects. Look at `bb-buck_top.png`: the only characters on the board are reference designators.

The intent *was* captured - J1 and J2 each carry a `Note` property reading "DC INPUT 18-30V, LEFT edge, wire opening off-board" / "DC OUTPUT 5V 2A, BOTTOM edge, wire opening off-board" - but both are `hide=yes` on F.SilkS, so they never plot and never reach the fab.

The only polarity indication that exists is each footprint's 0.3 mm pin-1 dot, and both are placed at the exact corner of the connector body outline: J1's at (26.370, 48.300) [0.30, 13.82], J2's at (43.690, 59.180) [17.62, 24.70]. Both are under the plastic body after assembly. So the assembled board has no polarity mark at all - this is the "polarity marks visible AFTER assembly" failure in its purest form.

J1 and J2 are the *same part*: KF128-5.08-2P, LCSC C474952, same size, same colour, same screw orientation. Nothing on the assembled board distinguishes the 24 V input from the 5 V output, or + from - on either.

This is not mode-excluded. requirements.md s2 requires "+ / -" on J1 and that J2 be "silk-labelled distinctly (VIN vs 5V OUT) so the two cannot be confused at the bench". s8 states outright that "the board has no reverse-polarity protection by mode, so the silk marking is the only defense against a swapped supply", and the constraints.json `placement.edges` note for J2 repeats it: "distinct silk (+/- vs 5V OUT) is the only defence against a swap." The mode removed the *protection*; it explicitly did not remove the *marking* - the marking is what the mode's own reasoning leans on.

Consequences, both real at the bench: reversed supply leads destroy U1 (accepted in s8 only because silk was assumed present); and landing the bench supply on J2 puts 24-30 V across a 25 V-rated output bank while back-feeding through U1's low-side body diode.

Fix: four short silk strings - `+/-` beside J1's two pins and `VIN` on its body, `+/-` beside J2's two pins and `5V OUT` on its body - placed so they sit *outside* the 10.16 x 10.5 mm connector body outlines and remain readable after assembly. There is clear board space above J1 (y_local 12-13.8) and left of J2.

2. Only 9 thermal vias under U1's exposed pad; the datasheet minimum is 12 (router) U1 EP at (38.970, 41.480) [12.9, 7.0], pad 2.5 x 3.5 mm.

The in-pad array is 3 x 3 at 0.85 mm pitch, 0.30 mm drill / 0.60 mm pad, centred on the EP. The LMR33630 datasheet Sec 10.1.1 is imperative and specific: *"Use a minimum 4 x 3 array of 10-mil thermal vias to connect the PAD to the system ground plane heat sink. The vias must be evenly distributed under the PAD."* That is 12 vias **under the pad**, and the board has 9 - 75 % of the stated floor.

The pipeline's own rule (`constraints.json _review_enforced (1)`, ">= 16 vias within 4.6 mm") *is* met: I count 18 GND vias within 4.6 mm of U1's centre. But that rule counts vias out in the surrounding F.Cu island, which reach B.Cu through a much longer lateral path than a via inside the pad. The datasheet's rule is about the pad, and it is the one that governs the joint's own heat path.

The 0.30 mm barrels partly recover the count: barrel-wall conductance scales roughly as n x d, so 9 x 0.30 = 2.70 against the datasheet's 12 x 0.254 = 3.05 - about 88 % of the minimum. Using the design's own 192 K/W-per-via figure, 9 vs 12 in parallel is 21.3 vs 16.0 K/W, ~5 K of extra junction rise at 0.92 W. That does not fail the part (Tj lands near 123 C against 150 C at 50 C ambient), but it eats most of the 2.06 C that `check_thermal` reports as margin - and `check_thermal` does not model the array at all, which is exactly why the constraints file made this review-enforced.

This is cheap to fix and it fits: a 3 x 4 array at the existing 0.85 mm pitch spans 1.70 x 2.55 mm inside a 2.50 x 3.50 mm pad, with 0.3 mm to every pad edge. Add one row (3 vias). See also warning 4 - adding vias makes the stencil question sharper, so fix the two together.

—

WARNINGS

3. TP1 sits on a dedicated switch-node spur, and the spur drags /SW into the FB/VCC corner (placement) TP1 pad at (43.970, 44.380) [17.9, 9.9]; the stub runs from inside L1's pad at (46.100, 43.000) to TP1, 2.54 mm of 1.52 mm-wide track.

`constraints.json placement.groups[probe]` is explicit: "TP1 must sit INSIDE the existing /SW copper, not on a spur, so it adds as little area as possible to the noisiest node." It does not. The existing /SW copper ends at y_local 9.45 (L1.1's pad edge); TP1 hangs 1.2 mm below it on a dedicated stub.

The area cost is modest - TP1's pad plus its stub is 4.74 mm2, 18 % of the 26.36 mm2 node - and both /SW ceilings still hold (26.36 mm2 against 40; 7.60 mm longest extent against 8). The real cost is spacing. Measured with and without the stub:

pair	/SW without the spur	/SW as built	factor
/SW -> /FB	2.930 mm	**1.038 mm**	2.8x closer
/SW -> /VCC	1.660 mm	**0.420 mm**	4.0x closer
/SW -> C7 pads (VCC bypass)	3.848 mm	**0.962 mm**	4.0x closer
/SW -> R1 pads (FB top)	4.566 mm	**1.948 mm**	2.3x closer

So a feature added purely for measurement convenience quadrupled the switch node's proximity to the gate-drive rail and nearly tripled it to the feedback node - the two nets `constraints.json` names as needing distance from /SW. The absolute coupling is small (tens of femtofarads; order 20 mV of per-cycle glitch on a 1 V reference through the FB node's own capacitance) and nothing here is illegal, but it is entirely avoidable: the U1.8 -> L1.1 corridor is 4.7 x 5.4 mm of existing /SW copper and a 1.5 mm pad fits inside it at roughly (44.97, 42.08) [18.9, 7.6], with TP2 following ~2.5 mm below. Deleting the stub also removes 2.5 mm of open-ended 1.52 mm track from the switch node.

4. U1's exposed pad is a single 100 % paste aperture with 9 open via barrels inside it (library) `aiee:ESOP-8_L4.9-W3.9-P1.27-LS6.0-TL-EP` pad 9: `(layers "F.Cu" "F.Mask" "F.Paste")`, 2.5 x 3.5 mm, no `solder_paste_margin` anywhere on the board.

Two uncontrolled quantities that do not reliably cancel:

- 8.75 mm2 of solid stencil aperture is ~1.05 mm3 of wet paste at a 0.12 mm stencil, roughly 0.5 mm3 of metal. IPC-7093 and TI's PowerPAD note (SLMA002) both call for a windowed/segmented aperture at ˜50-80 % coverage on thermal pads this size, precisely to stop the package floating on the EP and tilting the 0.6 mm gull-wing joints.

- The pad's mask aperture is 2.5 x 3.5 mm, which opens the mask over all 9 in-pad vias regardless of the board's `(tenting (front yes) (back yes))` setting. Nine 0.3 mm x 1.6 mm barrels hold ˜1.0 mm3 - about twice the available solder metal. Bottom-side tenting limits how much escapes, and TI's own guidance accepts unplugged holes at or below ˜0.33 mm, so this is a risk rather than a certainty - but it is unmanaged, and the EP is the entire heat path on 2 layers.

Fix: give pad 9 a windowed paste pattern (4-5 apertures, ˜70 % coverage) positioned between the via locations. Do this in the same pass as error 2.

5. L1 sits 0.50 mm from the top board edge (placement) L1 centre (52.670, 41.230) [26.6, 6.75]. Body 12.3 x 12.3 x 8.0 mm (SMDRI127-150MT / C40000 - note the footprint is *named* RLF12545T, which is a different part; the pads match the fitted part, so this is a library-naming issue, not a land-pattern error). Silk/body outline reaches y_local 0.55, courtyard 0.60, and the true 12.5 mm outline reaches 0.50 mm from the edge. See the top-right of `bb-buck_top.png` - the inductor is visually flush with the top edge.

JLCPCB's assembly guideline is >= 1 mm component-to-board-edge, and the depanel question matters more than the placement question: a 12.3 mm, 8 mm-tall part whose solder joints sit 0.5 mm from a V-scored edge is exposed to snap stress at depanelisation. Nothing else on the board is under 1 mm except the two terminals (intentionally, at 0.30 mm, openings at the edge) and C9 at 1.15 mm; C2/C3 are at 1.05 mm.

Fix: either move L1 >= 1 mm off the top edge (there is 12.1 mm of clear board below it) or specify routed/tab depanelisation rather than V-score in the P9 fab notes.

6. Both GND screw-terminal pins are flooded solid into full-board pours on both layers (plane) J1.2 at (31.670, 55.920), J2.2 at (51.310, 53.880). Both zones carry `(connect_pads yes (clearance 0.5))` - KiCad's *solid* pad connection. The `thermal_gap`/`thermal_bridge_width` values present in both zones are inert while `connect_pads` is `yes`.

Each of those pins is a 1.6 mm drill in a 2.4 mm pad, solidly tied to a 595.5 mm2 1 oz F.Cu pour on top and a 783.3 mm2 pour on the bottom - four solid plane connections to heat. requirements.md s7 explicitly allows the fallback path "hand-soldered after SMT", and that is where this bites: a normal iron will not get a plane-connected 1.6 mm barrel to reflow temperature reliably, and a cold joint on J2.2 is the 2 A return path. JLC's selective/wave process with preheat copes better, but the assembly route is not yet decided.

Fix: switch both GND zones to thermal-relief pad connection for through-hole pads (`connect_pads thru_hole_only` with the existing 0.5 mm gap / 0.5 mm spoke), or add per-pad thermal relief on J1.2 and J2.2 only. The SMD pads should stay solid.

7. The probe pair is mislabelled on the assembled board (silk) This is my triage of the four `check_silk` refdes-ambiguity warnings. All four are geometrically real - I reproduced each - and three are cosmetic. One is not:

- **R1 label at (45.698, 46.910), 0.10 mm from TP2** - escalate. TP2 is the scope ground pad of the one measurement feature the owner bought at A4, and TP2's *own* label sits at (45.470, 49.555) which is inside J2's courtyard, so it disappears under the terminal body at assembly. The result is that the assembled board labels the scope-ground pad "R1", while R1's actual body is 4.6 mm away. On a board whose stated purpose is to be measured, that is a usability defect, not a cosmetic one.
- C3 label at (47.583, 36.780), 0.68 mm from L1 - real, and it lands inside L1's courtyard so it also vanishes at assembly. Cosmetic; fix in the same pass.

- C7 label at (45.695, 45.030), 0.03 mm from L1's courtyard edge - real, cosmetic.
- J2 label at (55.525, 52.330), 0.53 mm from C4 - real, cosmetic.

No waiver recommended on any of the four: `place_edit.py move_text` is the cheapest fix on this list, and it is the same pass that has to add the silk from error 1.

Separately, I confirmed the owner-accepted count of hidden labels is exactly 7 and found nothing worse hiding under a body: TP2, C5, C8 and C9 under J1; C3 under L1; L1 under J2; J1 under its own body. The two that carry consequence are TP2 (above) and L1, which loses its identity entirely after assembly.

8. Three of the four output capacitors are 20-26 mm from the inductor, behind a 1.00 mm pour neck (placement) C4 (57.270, 49.380) is the only output cap near the converter, 8.2 mm from L1.2. C5/C8/C9 sit at x_local 14.2, y_local 15.6 / 19.1 / 22.6 - 20 to 26 mm from L1.2 - and reach +5V only through the priority-1 F.Cu zone bridge, whose narrowest section I measure at **1.002 mm**, at (40.970, 48.750) [14.9, 14.3].

`constraints.json placement.groups[output]` anchors all four on L1. As built the bank is split 1 + 3. At 400 kHz the ~20 nH of the run to the far three is ~50 mOhm against their own ~20 mOhm impedance, so roughly only C4 filters the fundamental ripple. The arithmetic still passes: ripple goes from ~8.3 mV (26 uF effective) to ~14.5 mV (C4 alone, ~15 uF after bias derating) against A3's 50 mV budget, and loop stability is untouched because at a ~40 kHz crossover 20 mm of trace is nothing, so all four caps count for Table 9-2's C_OUT requirement. The stated load is resistive, so there is no transient case to worry about either.

Reporting it because it is a silent deviation from the architecture with a 1.7x ripple cost, not because the board fails. If it is accepted, the +5V check_pdn waiver text should be amended (see waivers below).

9. The +5V pour neck is never measured by check_current (review) `check_current.py` reaches its `pour_neck` evaluation only for entries it treats as plane-carrying; on this board `GND` (with `plane_fed: true`) gets its pours eroded and measured, and `+5V` gets `min_track_mm_by_layer: {"F.Cu": 1.52}` and nothing else. The +5V zone is 18.26 mm2 of real current-carrying pour with a 1.002 mm neck and the checker never looked at it.

On the merits the neck is fine, and I want to be clear it passes rather than squeaking by: the 2 A DC load path is L1.2 -> 1.8 mm track -> J2.1 and never crosses the bridge. What crosses is the ripple share into three caps plus the divider's 39 uA - at most ~0.15-0.2 A against IPC-2152's ~1.6 A capability for 1.00 mm on 1 oz outer at dT 10, roughly 8x margin.

A second instance of the same blind spot, also benign: `pour_neck` uses *vias* as its attachment points, so the hot loop's F.Cu ground bridge between C1.2 and U1.1 - which has no vias by design, correctly - is never tested. I measured it at 0.935-1.230 mm continuous, against the 0.56 mm that 1.1 A needs. Fine, but untested.

Recorded so the hole is visible, not because either measurement fails.

10. The GND F.Cu override is anchored to a checker artifact, not to the copper (review) Audit of resolution 1. **The 0.6 A figure is honest - conservatively so.** I reproduced every number in the WO-3 note independently: B.Cu min neck 5.937 mm against the 1.520 mm requirement (3.91x, one unbroken 783.34 mm2 fill, zero B.Cu tracks); F.Cu min neck 0.322 mm; each of C2.2, C3.2, C4.2, C5.2, C8.2, C9.2, C10.2 carries 3 vias within 1.5 mm; U1's EP carries 9; both GND terminals are THT so the 2.6 A return enters B.Cu through a 1.6 mm barrel at J2.2 without touching an F.Cu neck. My own estimate of what actually crosses the 0.32 mm sliver is 30-50 mA - the sliver is ~20 mOhm against a ~0.7 mOhm via-B.Cu-via shunt, so it takes about 3 % of any current that wants to pass - i.e. 0.6 A is 12-20x conservative. The engineering is sound.

Two corrections to the record, and one durable risk:

- The note says the F.Cu pour "splits into 5 mutually-isolated LEAF-TAP via groups". It does not: the F.Cu GND fill is a **single connected region of 595.5 mm2** (I rasterised it at 0.02 mm and ran connected-component labelling - one component). The leaf-tap argument is about current *share*, and in that form it is correct; the word "isolated" is not.

- `pour_neck` reports its position as `parts[0].representative_point()` of the eroded fill. The override anchor (30.3575, 46.9464) sits in copper that is **7.56 mm wide** and about 9 mm from either true pinch at (40.550, 35.141) / (35.150, 35.141) - which are slivers along the *top* board edge, squeezed between the 0.5 mm pour inset and C2.1/C3.1's +VIN pads. So the override is placed on the checker's reporting artifact rather than on the defect.

- The margin is 0.36 mm. I confirmed the note's own measurements: r <= 1.8 mm captures 0 vias, and the nearest via-cluster centroid (C10.2's, at (32.05, 46.18)) is 1.858 mm away, so the default r = 2.0 would manufacture a new error. Any repour that moves the representative point by more than ~0.36 mm either breaks the fix or silently widens it. And because every future F.Cu GND neck on this net would be reported at approximately the same point, this single override will suppress them too, at any severity, without saying so.

Recommendation: keep the override (the physics is right), but re-anchor it on the real pinches at (40.550, 35.141) and (35.150, 35.141) with a radius that covers them and nothing else, and record that `pour_neck`'s reported position is not the neck.

11. /SW-to-GND clearance is 0.1275 mm, and the "no pair exceeds 30 V" premise is not true for /SW (review) Minimum measured at (45.545, 40.757) [19.475, 6.277], between the /SW track into L1 and the F.Cu GND pour. +VIN-to-GND has the same 0.1275 mm minimum at (34.052, 41.746). Both are the board's DRC floor, so this is a global pour-clearance setting rather than a routing slip.

The reasoning in `constraints.json _absent_keys` - "no net PAIR on this board exceeds 30 V (check_creepage's derived check engages only ABOVE 30 V)" - holds for every DC pair, but /SW is not a DC node. It swings 0 to VIN each cycle and overshoots on the turn-on edge; on a 2-layer board with a 2.3 mm2 loop, 1.2-1.5x VIN is the normal range, i.e. 36-45 V at the 30 V corner. That puts /SW-GND in IPC-2221's 31-100 V band, where external conductors under permanent polymer coating (B2) want 0.13 mm. The board has 0.1275 mm.

I am not claiming this board fails - 0.127 mm of masked FR-4 will not break down at 45 V, and this is standard JLC 2-layer practice. I am flagging that the stated basis for `check_creepage` being a clean no-op has a hole in it, and requirements.md s8 explicitly contemplates the owner lifting the A1 ceiling later, at which point that hole matters. Recording the switch-node overshoot as a known exception is the fix, not copper.

12. The GND void under L1 was never cut, and cutting it would break the 2-layer story (plane) `constraints.json _review_enforced (4)` requires "a GND void under the body" for L1. Measured over L1's 12.3 x 12.3 mm footprint: F.Cu GND covers 68.2 %, and over the core between the two pads it covers **99.5 %**; B.Cu GND covers 100 % of both.

Two things follow. First, the requirement was silently dropped - nothing in the pipeline enforces it and no rule area exists anywhere on the board. Second, it should probably stay dropped: the *only* mitigation this design offers for being 2 layers where the datasheet asks for 4 is an unbroken B.Cu pour under the hot loop, /SW and L1, and cutting a void under L1 on B.Cu would directly contradict it. The fitted part is a shielded/composite drum (SMDRI127-150MT), so copper beneath it is normal practice and the eddy-loss argument is weak.

Recommendation: waive on B.Cu permanently and amend the constraint to say so; optionally cut the F.Cu core void only, which costs nothing.

Related, same clause: L1's /SW-side copper is 26.36 mm2 against the ">= 40 mm2 of pad copper per terminal" in the same rule (the +5V side is 83.4 mm2 and passes). That rule cannot be met on the /SW side without breaching the "/SW <= 40 mm2" ceiling in clause (2) of the same file. Since L1 dissipates 0.27 W - below the 0.5 W threshold the architecture itself uses to decide a part needs a thermal entry - the EMI ceiling should win and clause (4) should be scoped to the +5V terminal only. No board change.

13. The declared 6.5 mm M3 keepouts do not exist (plane) `_review_enforced (3)` calls for hand-added KiCad rule areas giving a 6.5 mm mask/copper keepout at each M3 hole, because `planes_gen` cannot void. The board has **zero** keepout zones; both GND pours run to 0.5 mm from the board edge and up to both NPTH barrels.

I checked the consequence rather than just the omission, and it is nil: within a 3.25 mm washer radius of H1 (29.570, 37.980) and H2 (57.570, 55.980) there is no non-GND copper on either layer - no tracks, no pads, and the +5V zone is nowhere near. A metal M3 screw and washer therefore short top GND to bottom GND, which is harmless and arguably desirable on a bench article.

Recommendation: waive and close the constraint as satisfied-by-inspection rather than adding rule areas. Re-check if any future revision routes non-GND copper into a corner.

—

Verdicts on the three resolutions taken this phase

1. The GND `overrides` region at 0.6 A / r 1.5 mm - accept the number, re-anchor the region. Full audit in warning 10. Every measurement in the note reproduces exactly; my independent estimate of the current in the neck is 30-50 mA, so 0.6 A is conservative by more than an order of magnitude. The radius is safe today (nearest via cluster 1.858 mm away). What I object to is that the region is keyed to `parts[0].representative_point()` of an eroded polygon, sits in 7.56 mm-wide copper 9 mm from the actual pinch, and therefore has an unstated blast radius over future findings on the same net.

2. Fixing `check_silk.py` rather than waiving TP1/TP2 - correct, and I verified the geometry. Both TestPoint footprints carry `fp_circle` r 0.950 mm, stroke 0.120, `(fill no)` - a ring occupying 0.890 to 1.010 mm - against a 1.50 mm circular pad, i.e. radius 0.750 mm. Real clearance pad-edge to silk-inner-edge is **0.140 mm**. The old checker built the circle as a filled disc and so reported 100 % pad coverage; that was a genuine false positive and fixing the checker was the right call. One residual: 0.140 mm is inside typical silkscreen registration tolerance (+/-0.10 to 0.15 mm), so the ring may land on the pad in production. Both are bare probe pads and JLC clips silk over exposed copper anyway, so I am recording this rather than filing it.

3. The two 'check_pdn' waivers - both hold; amend the +5V reasoning.

- *GND*: sound without qualification. GND is a return net and cannot be decoupled to itself; the finding exists only because GND is deliberately declared as a power entry to keep pour necks at error severity, which is the right trade. Compensating controls verified by me: drc_routed 0/0, no GND error in check_current, and a B.Cu trunk measuring 5.937 mm minimum neck against 1.520 mm required on one unbroken 783.34 mm2 fill.

- *+5V*: the category argument is correct - an output bank hangs on no IC power pin, so `check_pdn` is looking for a structure that rightly does not exist. But the stated compensating control, "the 4 x 22 uF bank is verified present in the netlist and on the board", reads stronger than the layout supports: one of the four is near the converter and three are 20-26 mm away behind a 1.00 mm pour neck (warning 8), and the P4 SPICE bench gates the DC setpoint, not the bank's HF position. Keep the waiver, amend the wording so the record is accurate.

Verdicts on the remaining machine warnings

- **20 x 'insufficient_transition_vias' on GND - waive, all 20.** Every one is an isolated single stitching via judged against the whole 2.6 A budget (needs 7). The 2.6 A return does not pass through any of them: it enters B.Cu at J2.2's 1.6 mm THT barrel, and each capacitor's ground already has its own 3-via drop. This is exactly the case `plane_fed`'s advisory downgrade exists for. One note rather than a finding: the advisory at (43.970, 46.900) is TP2, the scope ground, on a single via - fine for a probe return, but a second via there is free.

- **1 x 'check_return_path' 'corridor_void' on /SW - waive.** 0.46 mm2 of deficit at (43.259, 34.891), with `crossing_len_mm: 0.00`. The polygon sits at y 34.84-34.98, i.e. inside the pour's own 0.5 mm edge inset at the top board edge. It is where copper stops on a 2-layer board, not a split, and no trace crosses it. The /SW corridor is otherwise 100 % covered - see below.

- **4 x 'check_silk' refdes ambiguity - do not waive, fix.** Triage in warning 7.

What I checked and found sound

Worth recording, because several of these are the load-bearing claims:

- **Both terminal openings face off-board, on different edges.** This was wrong once already, so I settled it two ways. Geometrically: J1 is at rot -90, which maps the footprint's wire-entry arrows (local +y) to global -x, putting the opening face 0.30 mm from the left board edge; J2 is at rot 0, putting its opening face 0.30 mm from the bottom edge. Visually: `bb-buck_edge-left.png` shows J1's two square wire openings face-on from the left edge, and `bb-buck_edge-bottom.png` shows J2's face-on from the bottom edge. Both correct, and on different edges as requirements.md s5 demands.

- **The 2-layer mitigation claim is true.** The B.Cu GND fill is a single unbroken 783.34 mm2 region with zero interior holes and zero disjoint pieces, and it covers 100.00 % of the hot loop, U1's body and EP, the /SW corridor (with and without the TP1 spur), L1's full footprint, the input caps and the output caps. There are no B.Cu tracks anywhere. `bb-buck_bottom.png` shows exactly that - bare pour, four THT pins, two mounting holes, the EP via array, nothing else.

- **The hot loop is excellent.** C1's +VIN pad is 1.651 mm centre-to-centre from U1.2 (0.550 mm pad edge to pad edge) and its GND pad is 1.651 mm from U1.1 PGND, on a 1.30 mm-wide direct track and an unbroken 0.935-1.230 mm F.Cu ground bridge respectively. The pads are uncrossed - C1.1 (+VIN, y 6.425) faces U1.2 (+VIN, y 6.360) and C1.2 (GND, y 5.025) faces U1.1 (PGND, y 5.090). Enclosed loop 1.65 x 1.40 mm = 2.31 mm2, matching the design intent. C1.2 correctly has *no* vias within 2.5 mm - the loop stays on F.Cu with B.Cu as its image plane, which is right.

- **/SW containment holds.** 26.36 mm2 against the 40 mm2 ceiling, longest extent 7.60 mm against 8 mm, F.Cu only, zero layer transitions, 4.05 mm from the nearest board edge and 8.7 mm from J2's nearest pad.

- **U1's pinout is wired correctly**, including pin 3 EN tied to +VIN (the datasheet states EN "can be connected directly to VIN" with an abs-max of VIN + 0.3 V) and pin 4 PG left open, which the datasheet permits.

- **W2 is satisfied**: R2's ground return is a dedicated 1.52 mm track from (37.270, 46.600) to (38.900, 43.100), landing inside the exposed pad, i.e. on AGND. The board-wide F.Cu GND pour makes the Kelvin intent partly notional, but check_irdrop puts the worst GND drop at 0.609 mV, which is 0.06 % of the 1 V reference - negligible.

- **The A3 amendment is on the board**: R1 = 102k, R2 = 25.5k, ratio exactly 4.000, 5.000 V nominal.

- **The 5 V feedback divider is well separated**: /FB to /SW 1.038 mm even with the TP1 spur, /FB guarded by GND pour on both sides, R1/R2 4.6 mm clear of L1's courtyard.
- **No courtyard overlaps anywhere**; every interface requirements.md names is present (J1, J2, TP1 + TP2 and nothing else); two M3 holes diagonally opposite, which requirements.md s5 explicitly sanctions in place of four.
- **Fiducials**: not required. Finest pitch on the board is 1.27 mm (SOIC-8) and JLC assembles from panel fiducials at that pitch. Not a finding.
- **P9 has not run** - `boards/bb-buck/fab/` is empty - so the depanel note in warning 5 and the stencil change in warning 4 both still land in front of the fab package rather than behind it.

Documentation drift (no board change)

`architecture/blocks.md` is stale against the built board in four places: it lists H1..H4 (the board has H1/H2, correctly, per requirements s5 and the constraints keepout note), C1 as 100 nF (built as 220 nF, which is what the LMR33630 layout section asks for), the output bank as 2 x 22 uF 1206 (built as 4 x 22 uF 1210 per the P4 fix), and omits C7, C8, C9 and C10 entirely. All four changes are recorded in `constraints.json` and were owner-approved; only blocks.md was not refreshed.

Waivers recommended

1. `check_current` `insufficient_transition_vias` x20 on GND - isolated stitching vias on a plane-fed net; the 2.6 A return enters B.Cu at J2.2's THT barrel and every capacitor ground already has 3 vias. 2. `check_return_path` `corridor_void` on /SW - 0.46 mm2 at the pour's own 0.5 mm edge inset, 0.00 mm of trace crossing. 3. Finding 13 (M3 keepouts absent) - verified consequence-free; close the constraint as satisfied-by-inspection.

Findings 8 and 12 are also reasonable waiver candidates if the owner accepts the arithmetic; both are recorded as warnings so the decision is explicit rather than silent.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake - digest (2026-08-15)

- bb-buck:  non-isolated buck, 18-30 V DC in (24 nom) -> 5 V @ 0-2 A out, screw
terminal both ends, bench supply + resistive load.
- Mode **ultra-bare-bones** recorded as a decision:  buck block +
datasheet-required
support only.  Defaults applied (qty 5, JLC PCBA single-sided, 0-50 C bench, no
enclosure, fewest honest layers, smallest honest outline, stop at P9).
- requirements.md by 'requirements-analyst'; 'check_requirements.py' PASS 9/9,
0
violations, before and after fold-in.
- Section 8 closed with NO safety question open, derived:  bench source (no
battery),
resistive load (no motor), ~2.5 A peak (< 3 A), 30 V is not >30 V and < 60 V
ES1.
```

```
- 4 questions asked in ONE batch, owner took all four defaults -> now
requirements:
A1 30 V hard max operating, part abs-max >= 36 V (headroom in the PART, no
clamp);
A2 no live hot-plug (drops the ~60 V ring case); A3 +/-3 % DC over full
line+load,
<= 50 mV pk-pk; A4 one switch-node probe pad + adjacent GND pad, nothing else.
- Left open for P1/P2:  sync vs async rectification; 2L vs 4L (2L assumed,
revisable
on thermal grounds at ~0.8-1.8 W / 50 C ambient).
- Fable tier out of credits -> fable/high spawns run opus/xhigh (ledgered).
```

P1

```
# P1 Research - digest (2026-08-15)

- Roster:  power-architect + component-scout(buck-regulator) only.  No
interface-spec
(screw terminals bind no standard), no reference-design (P3 part-level coverage
research produces the vendor layout digest for the CHOSEN IC instead).
- Topology ruled **synchronous integrated-FET, ~500 kHz (350-600), L = 10 uH**.
Async
rejected:  +0.42 W (+33 %) plus a 0.77 W diode at Tj ~126 C vs a 125 C rating
(D=0.17
means it conducts 83 % of every cycle at 30 V). >= 1 MHz rejected:  1.58-1.93 W
and
t_on(30 V) collides with min-on-time.
- Part envelope for P3:  Vin abs-max >= 36 V, **Rds_LS <= 85 mOhm (rank on LS
not HS+LS;
at D=0.167 the LS FET conducts 5x longer)**, t_on_min <= 130 ns, Tj 150 C,
exposed pad
mandatory, P_U1 <= 0.95 W at 30 V/2 A. A 3 A-class part, not a marginal 2 A
one.
- **2L CONDITIONAL** - the screen passes by 2 C (0.92 W x 73.8 C/W = 67.9 vs
dt_c 70).
Needs sync + Rds_LS <= 85 mOhm + outline >= ~1000 mm^2 + >= 16 thermal vias +
Tj 150 C.
Layer count FOLLOWS THE PART; re-run at P3 before P5 fixes the stackup.  Mode
note:
"smallest honest outline" has a numeric floor - the outline IS the radiator.
- Scout, 5 candidates, every figure datasheet-sourced:  top **LMR33630ADDAR**
C841384
(sync, 38 V, 400 kHz, $0.74@5, stock 6885); AOZ1284PI C48060 async 40 V $0.85;
TPS54560BDDAR C1850354 65 V $1.06.  AP63357Q out at 35 V abs-max (1 V short of
A1);
LMR33630's 1.4/2.1 MHz siblings pulse-skip at 30 V - only the 400 kHz "A"
clears it;
XL1509 is the only Basic part but its own +/-4 % already breaks A3's +/-3 %.
- Convergence:  the scout's top pick sits inside the architect's envelope.
```

P2

```
# P2 Architecture - digest (2026-08-15)

- One block (B1 sync buck) + J1/J2 screw terminals + TP1/TP2 SW probe pair + 4x
M3.
14 parts, 6 nets, ONE FLAT ROOT SHEET. Canonical nets:  +VIN /SW +5V GND.
Stackup JLC2313_1.6 2L 1 oz CONDITIONAL (trigger:  P_U1 > 0.95 W = Rds_LS >
~110
mOhm at 400 kHz).  **Outline 40 x 30 mm FINAL** - owner-confirmed at H1.
```

```
- fsw 400 kHz / L 15 uH. P1's fixed-output preference LOST to stock reality ->
adjustable + 0.1 %/25 ppm divider, closing A3 on worst-case SUM. lint 0/0.
- Architect corrected 3 briefing premises, all verified (rules_gen already
buckets
per IPC-2152 width; planes[] REPLACES defaults and rejects ‘_note‘; pdn:false
also
skips check_irdrop) -> LEARNINGS.md entry, incl.  a latent sbuck-5v3a defect.
- Coverage at the mode floor:  1 slot / 10 classes all gap (8 maturity-only, 3
substantive).  Mapper skipped as a provable no-op.  Research block-buck-1:  3
TI
sources, 12+ pages read visually, 9 records.  Reader REFUTED 1 (record said 0.3
uH,
figure prints **9.3 uH** - 31x, mis-sized the damping criterion); corrected,
re-read fresh -> **9 verified / 0 refuted**.  Re-run:  9 covered / 1
provisional
(inrush) / 0 gap.
- Design-relevant:  hot-plug at 30 V rings to 45-49 V vs 38 V abs-max (A2 is
what
makes the clamp-free mode safe); the EP is **AGND**, so the via array is a
loop-accuracy item; 2L is a documented deviation from the datasheet's layout
rule 5
at ~5 dBuV/m -> P6/P7 must keep the bottom pour unbroken under hot loop, SW and
L.
- H1 APPROVED. Promotion of the 9 records deferred to run close with bring-up.
```

P3

```
# P3 Parts + Library - digest (2026-08-15)

- parts.json:  10 distinct parts / 13 refdes, ~$1.63/board, 1 Basic + 9
Extended.
U1 has NO drop-in alternate (async candidates D1-rejected; frequency siblings
cascade into L/C_OUT re-derivation).
- **D4 MET - 2 LAYERS CONFIRMED**:  real Rds_LS 66 typ / 110 max mOhm over full
temp
-> P_U1 0.57 typ / 0.76 W worst-case vs the 0.95 W gate, ~20 % margin (the
placeholder model predicted 2 C). D3 met (15 uH + 2x22 uF inside both windows).
D11 met (Isat 8 A vs 6.6 A floor, DCR 27 mOhm, 12.3 mm body).
- **A3 AMENDED by owner**:  <= 50 mV holds 200 mA - 2 A, PFM burst ripple
accepted
below.  DDA has no MODE pin, auto-mode only (8.4.1).  DC accuracy unchanged.
- Extraction validated (9 pins, EP = **AGND**).  Traps:  the 68 ns min-on-time
headline
is the VQFN part's (DDA 75-108 ns vs our 417 ns need); 10.1.1's 4x3=12 vias
contradicts Fig 10-2's ~6 - our >= 16 governs.  Two DDA revs with different EP
sizes.
- Library 10/10 pulled, 6/6 load-check, 27 pins retyped, no B.Cu lands, no
netless PTH
in the EP (LM5017 precedent did not repeat).  RULED: accept the vendor U1 land
(warning-level; it is what JLC's process is built around) and the library-wide
courtyard-cuts-pads class (placelib unions pad bbox).
- **Coverage P3:  1 covered (part slot) / 1 provisional (inrush) / 0 gaps.**
- OPEN RISK to P4 review:  C_OUT 2x22 uF vs TI's 4x22 uF reference - stability
or only
ripple?
```

P4

```
# P4 Schematic - digest (2026-08-15)
```

- ONE schematic agent (flat root sheet, recorded deviation). Generator
‘kicad/gen/root.py‘ is the source; the .kicad_sch is build output.
- **Gates: erc 0 errors / 0 warnings; netlist_audit exit 0; sim PASS
(0 err / 0 warn, 10 bounds, 18 measures).** 22 components, 8 nets.
- Cleared by the reviewer on datasheet evidence: EN->+VIN direct (Table 6-1
sanctions it, and EN abs-max is VIN+0.3 so it cannot be violated by
construction), PG open no-connect, EP=AGND tied to GND (the datasheet
REQUIRES the tie), CFF correctly open (confirmed three ways), full abs-max
sweep at 30 V, ton 417 ns vs 108 ns, Isat 8 A vs the strict 5.05 A criterion.
- **Fix pass, 3 changes.** E1 (error): C_OUT 2 -> **4 x 22 uF** - 2x is
outside
the internal compensation's optimised range (Sec 9.2 p19, quantified only in
Table 9-2 p20 whose 400 kHz/5 V row is 4x; every 400 kHz row is 4x). Failure
would have been SILENT at DC, visible only on load/line steps. W1: input bulk
2 -> **3 x 10 uF** (the 10 uF minimum is EFFECTIVE, and 2x fell to ~8.5 uF in
the worst corner). A3: divider **100k/24.9k -> 102k/25.5k**, exact 4.000
ratio, 5.000 V nominal. +$0.52/board.
- **A3 amended a second time and requirements.md updated.** Sec 7.7 gives
-1.5/+2.5 % regulation for IOUT 0 A to max (vs +/-1.5 % above 1 A) - the same
PFM root cause as the ripple carve-out. Stacked worst case was 5.161 V vs the
5.15 V ceiling; recentring moved it to 5.144 V. My earlier "DC accuracy
unaffected" statement was wrong and is retracted in requirements.md.
- Sim calibration proved the light-load warning bound is load-bearing: a
revert
to 100k/24.9k produces ZERO errors and is caught ONLY by ‘vout_ll0a‘. All four
single-E96-step mutations trip at error severity.
- constraints.json placement.groups repaired by the orchestrator (hotloop was
missing C10, output was missing C8/C9 and still said "1206"); W2 carried into
the feedback group note - R2's return must pin to the AGND/EP reference.
- Library: C861068 has NO EasyEDA CAD record at all (lib_pull reports that
identically to a rate-limit) - symbol cloned from its same-family sibling.

P5

P5 Board Setup - digest (2026-08-15)

- **Outline CUT 40x30 -> 35x25 mm (1200 -> 875 mm2, -27 %) on owner
direction**
("priority is a tight layout"). Defensible because the facts moved: H1's 1200
mm2
rested on the PLACEHOLDER 0.92 W vs a 0.95 W gate, and the real part came in at
0.76 W worst-case. 875 mm2 keeps the pour above check_thermal's ~645 mm2
saturation knee, so 0.76 W still gives ~56 C rise against the 70 C limit.
- **Mounting holes 4 -> 2 diagonal** (H1 bottom-left, H2 top-right). 4 x M3
with
6.5 mm keepouts would claim ~170 mm2 of 875 - 19 %, all in corners the hot loop
and output bank need. requirements.md s5 sanctions exactly this.
- Generator edited for the hole count; schematic regenerated; **erc re-gated
0/0**.
- CORRECTED an owner premise, verified by grep not by citing the doc:
‘--outline‘
exists ONLY on board_init.py and no other script writes Edge.Cuts, so the edge
cut CANNOT be moved after placement - a resize rebuilds from the netlist and
discards placement and routing. Strategy recorded instead: place -> measure
->
resize + RE-PLACE if there is slack -> route ONCE at the final size.
- board_init: **parity 0** (the first run's 8 parity errors were duplicate
H1-H4 -
the schematic already carried them and --mounting-holes added a second set;
flag

dropped). 20 components, 8 nets, outline_bbox 35.00 x 25.00, profile
2layer_1oz.
33 unconnected = expected pre-routing. ONE residual error: L1 pad 2 hangs
over
the bottom edge in the shelf-pack dump - a row-packer artifact, not a
feasibility
verdict; P6 replaces the entire placement and the 'place' gate confirms.
- rules_gen: 12 rules, capability_class 2layer_1oz, **netclasses split per
IPC-2152 width** - Pwr_0p56mm (+VIN, 1.1 A) and Pwr_1p52mm (/SW, +5V, GND,
2.6 A). No flattening; no HV rules (30 V does not trip creepage).
- Separation J2/J1-to-L1 relaxed 8 -> 6 mm: the 8 mm was explicitly justified
as
"what a 40x30 outline can honestly give", i.e. outline-derived, not physical.

P6

P6 Placement - digest (2026-08-16)

- **place gate PASS 0/0** (all five legs ran); kc.py drc clean but for the 33
expected pre-routing unconnected; silk_place 0 residual. hpwl 306.7 -> **125.0
mm**.
Route probe **completion 1.00** (33/33 bare, 19/19 with pours, re-verified by
importing the session and re-running DRC).
- **SCRAP AVERTED: both provisional connector rotations were wrong** - J1 0 ->
270,
J2 90 -> 0. The WRL carries (rotate 0 0 180), the documented coincidence trap,
so
the agent resolved native entry on a 3D test board and confirmed on
orthographic
left/front renders. As declared, BOTH wire openings faced inward = scrap.
- Hot loop hand-placed and **locked before anneal**: C1 220 nF 1.72 mm from
BOTH
VIN(2) and PGND(1), pads UNCROSSED, ~2.2 mm2. A rotation-sign error that
mirrored
C1/C6 into a self-crossing loop was caught and fixed mid-run.
- **All annealer candidates rejected on structure** (each put the FB divider
against
L1's SW pad and TP1/TP2 outside the switch-node copper - what the cost function
cannot see). The hand placement also won on hpwl, 125.0 vs 130.3-130.8.
- Cin/Cout ground separation 10.49 mm (record wants 1-2 cm). W2 satisfied by
geometry: R2's GND pad ~4.1 mm on a clear straight path to the EP/AGND island.
TP1 inside the /SW copper; /SW ~30 mm2 vs the 40 mm2 ceiling. Bottom pour
unbroken under hot loop, /SW and L1 (only J1/J2 THT pads + 2 M3 holes
penetrate).
- **OUTLINE FINAL AT 35x25, measured**: occupied 34.90 x 24.90, 0.05 mm slack
all
round, fill 72.7 %, largest inscribed empty rectangle 4 x 4 mm - one more 1210
does not fit anywhere. Best conceivable shrink 34x24 (-7 %) would eat the
0.15-0.40 mm gaps this layout runs on. Owner's place-measure-resize plan
ANSWERED:
no resize. H3 approved; 7 refdes under bodies accepted (no free silk area).
- **Toolchain: fixed routelib's score regex** (crashed route_auto ON SUCCESS -
the
no-unrouted log line ends in a full stop that '[\d.]+' swallowed). Also
recorded
two silently-non-enforcing constraint mechanisms: keepouts are never
translated
from board-local, and separation is centre-to-centre and skipped when either
ref

is locked - all four declared pairs here were unchecked. Verified by hand
instead.
- Carried to P7: planes_gen made only **10** EP thermal vias vs our >= 16
(datasheet
text minimum 12; its Fig 10-2 draws ~6 - do not copy the figure).

P7

P7 Routing - digest (2026-08-16)

- **drc_routed PASS, 0 violations (err+warn, parity + all track errors,
refilled);
completion 1.00 (33/33 ratsnest, 0 unconnected_items).**
- **route_critical REVERTED and re-done by hand.** KRT put /SW, +VIN and +5V
copper
on B.Cu, slicing the ground shield through the hot loop and the switch node -
the
one thing recovering this board's 2L deviation. Restored from git, routed
deliberately: 36 tracks, **ALL F.Cu, zero B.Cu tracks board-wide**, zero vias
on
+VIN / /SW / +5V.
- **route_auto NOT run (accepted deviation)**: the board already measured DRC
0 /
completion 1.00, and exporting a DSN of a fully-coppered board is the
documented
Freerouting wedge case whose KRT finish can ADD B.Cu - the failure just
reverted.
route_cleanup skipped per the standing 2L-pour advice.
- Binding constraints, all MEASURED not asserted: B.Cu GND one unbroken 783.3
mm2
island with 100 % coverage under U1, /SW and L1; **18 GND vias within 4.6 mm of
U1**
(9 in-pad + 7 ring + 1 grid) vs the >= 16 requirement; dedicated 1.52 mm AGND
spine
R2.2 -> EP that never touches PGND (W2 closed); +5V under J2 with **0.0000
mm2**
inside H2's TRANSLATED washer rect; /SW 27.02 mm2 vs the 40 mm2 ceiling, F.Cu
only.
- +5V reaches the left cap bank through an 18.3 mm2 pour bridge (neck 1.00 mm)
- the
sanctioned pour-fan-in remedy, since placement split the bank and the
R1.2/TP2/C5.2
pinch caps any track at ~0.95 mm. DC load path untouched at 1.8 mm; only cap
ripple
crosses, and FB senses on the no-DC side.
- Two new tool findings (workspace LEARNINGS): planes_gen's default
THERMAL-RELIEF pad
connection stranded U1's PGND on a 1.2 mm2 island in a 0.79 mm corridor - fixed
with
'connect: solid' via a planes-only sidecar; and pad extents must come from
'geom.pads_of().poly.bounds', since L1's land is 2.9 x 5.4, not the raw (size
5.4 2.9).
- Carry-forwards to P8 (not blockers): +VIN hot-loop widths geometry-capped at
1.30 mm
(C1->VIN) and 0.66 mm (bulk feed) vs the 1.5 mm wish - both above the 0.56 mm
floor;
H1/H2 washer keepouts still carry GND pour (planes_gen has no void support, all
copper and vias are clear); plane_repair mislabels the +5V bridge a dead_island.

P8 *(no digest recorded)*

Gate attempt history

(no entries)

Issues

Id	Phase	Kinds	Severity	Status
1	P8	silk_over_pad	error	fixed
2	P8	pdn_undecoupled	error	open
3	P8	pour_neckdown	error	fixed

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-15T18:56:25	Architecture approved. Outline 40x30 mm CONFIRMED by owner as final (irreversible at P5 board_init). Record promotion deferred to run close with bring-up evidence - verified workspace records still feed P3/P6/P7 spawns meanwhile; inrush stays provisional until then.
2	P4	approved	2026-08-15T22:34:04	Schematic approved: erc 0/0, netlist_audit 0, sim pass. Owner ruling on placement pressure: NOTHING gives - if the 7 x 1210 bodies cannot fit while placing the hot loop honestly on the 40x30 outline, STOP and report; do not relax the hot-loop grouping and do not move parts to the bottom side.
3	P6	approved	2026-08-16T01:35:58	Placement approved. Silk: owner accepted 7 refdes labels under bodies (C2,C3,C5,C8,C9,L1,TP2) - at 72.7% fill with a 4x4 mm largest empty rectangle there is nowhere to move them; TP1's label was already hand-moved out because it marks the switch-node probe. Outline CONFIRMED final at 35x25 by measurement (34.90x24.90 occupied, 0.05 mm all round) - no resize, route here.
4	P8	not reached		
5	P10	not reached		

10 Artifact Index

File	Size	Note
`state.json`	66,774 B	
`kicad/bb-buck.kicad_pcb`	159,595 B	
`kicad/bb-buck.kicad_sch`	81,019 B	
`brief/brief.md`	605 B	
`requirements.md`	18,721 B	
`architecture/blocks.md`	10,091 B	
`architecture/constraints.json`	21,873 B	
`architecture/decisions.md`	9,227 B	

File	Size	Note
architecture/power_tree.md	10,485 B	
architecture/sheets.md	7,162 B	
architecture/stackup.md	7,324 B	
reports/board_init.json	2,765 B	
reports/check_current.json	17,246 B	
reports/check_irdrop.json	5,013 B	
reports/check_pdn_z.json	1,033 B	
reports/check_silk.json	3,420 B	
reports/gate-drc_routed.json	498 B	
reports/gate-erc.json	688 B	
reports/gate-place.json	989 B	
reports/gate-sim.json	455 B	
reports/gate-verify.json	2,671 B	
reports/place-edit.json	3,812 B	
reports/place-final.ops.json	3,077 B	
reports/place.json	4,347 B	
reports/review-board.json	15,202 B	
reports/review-board.md	28,819 B	
reports/review-schematic.json	2,943 B	
reports/review-schematic.md	24,242 B	
reports/rules_gen.json	1,455 B	
reports/schematic.pdf	76,058 B	
reports/silk_place.json	2,770 B	
reports/sim.json	1,214 B	
reports/top.net	34,564 B	
reports/verify-waivers.json	3,445 B	